Complete characterization of Maximum Pairwise Cross Intersecting Families

Yang Huang* Yuejian Peng†

Abstract

Two families $\mathcal{A}$ and $\mathcal{B}$ are cross intersecting if each member of $\mathcal{A}$ intersects each member of $\mathcal{B}$. For integers $n, k_1, \ldots, k_t$ with $k_1 \geq \cdots \geq k_t$, we say that a tuple $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an $(n, k_1, \ldots, k_t)$-cross intersecting system if $\emptyset \neq \mathcal{F}_i \subseteq \binom{[n]}{k_i}$ for each $i \in [t]$, and $\mathcal{F}_i$ and $\mathcal{F}_j$ are cross intersecting for any $i \neq j$. Let $M(n, k_1, \ldots, k_t)$ be the maximum sum of sizes of families of an $(n, k_1, \ldots, k_t)$-cross intersecting system. About fifty years ago, Hilton and Milner [14] determined $M(n, k, k)$ under the condition $n \geq 2k$. Later, Frankl and Tokushige [9] obtained $M(n, k_1, k_2)$ for $n \geq k_1 + k_2$. Recently, Shi, Frankl and Qian [21] determined $M(n, k, \ldots, k)$ for $n \geq 2k$. Extending this result, Huang, Peng and Wang [11], as well as Zhang and Feng [22] independently determined $M(n, k_1, \ldots, k_t)$ for $n \geq k_1 + k_2$.

All aforementioned results require the condition $n \geq k_1 + k_2$. The goal of our paper is to weaken this to the most natural constraint $n \geq k_1 + k_t$. In fact, in the range $n < k_1 + k_t$, if $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an extremal $(n, k_1, \ldots, k_t)$-cross intersecting system, then $\mathcal{F}_1$ is the family of all k_1-element subsets of $[n]$ since any k_1-subset intersects with any k_i-subset for $i \geq 2$ when $n < k_1 + k_t$. So determining $M(n, k_1, k_2, \ldots, k_t)$ reduces to determining $M(n, k_2, \ldots, k_t)$. Therefore, $n \geq k_1 + k_t$ is the most natural constraint.

In the range $n < k_1 + k_2$, any family $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}$ and any family $\mathcal{F}_2 \subseteq \binom{[n]}{k_2}$ are clearly (freely) cross intersecting. This leads to no constraint on the pair of $\mathcal{F}_1$ and $\mathcal{F}_2$. Similarly, for smaller integer n, e.g., $n < k_1 + k_3$, we shall get more pairs of families that have no intersection constraint. For such pairs of families, their sizes are independent of each other. Since $n \geq k_1 + k_t$, every family of $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ must be constrained by some other families. Roughly speaking, we deal with weaker intersection constraint on the families $\mathcal{F}_1, \ldots, \mathcal{F}_t$. The interplay between families of $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ becomes highly uncertain. This difference changes the feature of this problem and causes considerable difficulties. In a previous paper [13], the present authors made the first step toward determining $M(n, k_1, \ldots, k_t)$ in the range $n \geq k_1 + k_3$ and established some foundational results. In this paper, we give a complete solution of determining $M(n, k_1, \ldots, k_t)$ in the full range $n \geq k_1 + k_t$ and characterize all extremal families.

Key words: Cross intersecting families; Extremal finite sets
2020 Mathematics Subject Classification. 05D05, 05C65, 05D15.

1 Introduction

A large variety of problems and results in extremal combinatorics deal with estimates on the sizes of families of sets with some restrictions on the intersections of their members. Let $[n] = \{1, 2, \ldots, n\}$ be the standard n-element set. For $0 \leq k \leq n$, let $\binom{[n]}{k}$ denote the family of all k-subsets of $[n]$. A family $\mathcal{A}$ is k-uniform if $\mathcal{A} \subseteq \binom{[n]}{k}$. We say that $\mathcal{A}$ is *intersecting* if any two sets of $\mathcal{A}$ have non-empty intersection. In the vast body of literature in the intersection

*School of Mathematics, Hunan University; Moscow Institute of Physics and Technology; E-mail: 1060393815@qq.com.

†School of Mathematics, Hunan University; E-mail: ypeng1@hnu.edu.cn. The research is supported by National Natural Science Foundation of China (No. 12571363) and National Natural Science Foundation of Hunan Province (Grant No. 2025JJ30003).

problem, one celebrated result is the Erdős–Ko–Rado theorem [5], which states that for $n \geq 2k$, every intersecting family $\mathcal{A} \subseteq \binom{[n]}{k}$ has size at most $\binom{n-1}{k-1}$. This is the best possible as witnessed by a k-uniform full star. The Erdős–Ko–Rado theorem has many interesting generalizations. Since then, various intersection theorems were established in the literature, and some of them were applied in other mathematical fields. For example, the Ahlswede–Khachatrian theorem [1], which determines the maximum size of $\mathcal{A} \subseteq \binom{[n]}{k}$ where any two sets intersect in at least t elements, plays a crucial role in the hardness-of-approximation theorem for the 'vertex cover' problem, proved by Dinur and Safra [3] (see [18]).

Two families $\mathcal{A}$ and $\mathcal{B}$ are called *cross intersecting* if $A \cap B \neq \emptyset$ for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$. For integers $n, k_1, \ldots, k_t$ with $k_1 \geq \cdots \geq k_t$, we say that a tuple $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an $(n, k_1, \ldots, k_t)$-cross intersecting system if $\emptyset \neq \mathcal{F}_i \subseteq \binom{[n]}{k_i}$ for each $i \in [t]$, and $\mathcal{F}_i$ and $\mathcal{F}_j$ are cross intersecting for any $i \neq j$. Let $M(n, k_1, \ldots, k_t)$ be the maximum sum of sizes of families of an $(n, k_1, \ldots, k_t)$-cross intersecting system. We say that an $(n, k_1, \ldots, k_t)$-cross intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is *extremal* if $\sum_{i=1}^{t} |\mathcal{F}_i| = M(n, k_1, \ldots, k_t)$.

In 1967, Hilton and Milner [14] determined $M(n, k, k)$ for $n \geq 2k$. To the best of our knowledge, the Hilton–Milner theorem was the first result on the sum of sizes of cross intersecting families. In 1992, Frankl and Tokushige [9] obtained $M(n, k_1, k_2)$ for any $n \geq k_1 + k_2$. Recently, Shi, Frankl and Qian [21] determined $M(n, k, \ldots, k)$ for $n \geq 2k$. Extending this result, Huang, Peng and Wang [11], as well as Zhang and Feng [22] independently determined $M(n, k_1, \ldots, k_t)$ for $n \geq k_1 + k_2$.

Theorem 1.1 ([11, 22])**.** *Let $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \mathcal{F}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ be non-empty pairwise cross intersecting families with $t \geq 2$, $k_1 \geq k_2 \geq \cdots \geq k_t$. If $n \geq k_1 + k_2$, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \max \left\{ \sum_{i=1}^{t} \binom{n-1}{k_i - 1}, \ \binom{n}{k_1} - \binom{n-k_t}{k_1} + \sum_{i=2}^{t} \binom{n-k_t}{k_i - k_t} \right\}.$$

Moreover, the extremal families are characterized as follows.

(i) For $n > k_1 + k_2$ or $n = k_1 + k_2 > k_1 + k_t$, the equality holds if and only if either there is a set $T \in \binom{[n]}{k_t}$ such that $\mathcal{F}_1 = \{F \in \binom{[n]}{k_1} \colon F \cap T \neq \emptyset\}$ and $\mathcal{F}_j = \{F \in \binom{[n]}{k_j} : T \subseteq F\}$ for every $j \in [2, t]$; or there exists $a \in [n]$ such that $\mathcal{F}_j = \left\{F \in \binom{[n]}{k_j} \colon a \in F\right\}$ for every $j \in [t]$.

(ii) For $t = 2$ and $n = k_1 + k_2$, the equality holds if and only if $\mathcal{F}_1 = \binom{[n]}{k_1} \setminus \{[n] \setminus A \colon A \in \mathcal{F}_2\}$.

(iii) For $t \geq 3$ and $n = k_1 + k_2 = k_1 + k_t$, if $k_1 > k_2$, then the equality holds if and only if $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t$ are full stars with the same center; if $k_1 = k_2$, then the equality holds if and only if $\mathcal{F}_1 = \mathcal{F}_2 = \cdots = \mathcal{F}_t$, and $\mathcal{F}_1$ is any intersecting family with size $\binom{n-1}{k_1-1}$.

For two families $\mathcal{A} \subseteq \binom{[n]}{k}$ and $\mathcal{B} \subseteq \binom{[n]}{\ell}$, if $n < k + \ell$, then $\mathcal{A}$ and $\mathcal{B}$ are clearly cross intersecting. Throughout this paper, such a pair of families is called *freely* cross intersecting. On the other hand, $\mathcal{A}$ and $\mathcal{B}$ are called *non-freely* cross intersecting if $n \geq k + \ell$, and $\mathcal{A}, \mathcal{B}$ are cross intersecting. If an $(n, k_1, \ldots, k_t)$-cross intersecting system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ contains at least two families that are freely cross intersecting, and also it contains at least two families that are non-freely cross intersecting, then we say that the system $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is of *mixed type*. Otherwise, we say that it is of *non-mixed type*.

The aforementioned references addressed the intersection problem exclusively for non-mixed type systems, which require the condition $n \geq k_1 + k_2$. It would be interesting to extend this study to mixed type systems under the condition $n < k_1 + k_2$. In a previous work [13], we established the first result for such mixed type systems in the range $k_1 + k_3 \leq n < k_1 + k_2$.

Theorem 1.2 (Huang–Peng [13]). *Let $t \geq 3$, $k_1 \geq k_2 \geq \cdots \geq k_t$ and $k_1 + k_3 \leq n < k_1 + k_2$. If $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \mathcal{F}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ are non-empty pairwise cross intersecting families, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \max\left\{\sum_{i=1}^{t}\binom{n-1}{k_i-1}, \sum_{i=1}^{2}\left(\binom{n}{k_i} - \binom{n-k_t}{k_i}\right) + \sum_{i=3}^{t}\binom{n-k_t}{k_i-k_t}\right\}.$$

We also determined all extremal systems that achieve the upper bound in [13]. The goal of our paper is to weaken this to the most natural constraint $n \geq k_1 + k_t$. In fact, in the range $n < k_1 + k_t$, if $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an extremal $(n, k_1, \ldots, k_t)$-cross intersecting system, then $\mathcal{F}_1$ is the family of all k_1-element subsets of $[n]$ since any k_1-subset intersects with any k_i-subset for $i \geq 2$ when $n < k_1 + k_t$. So determining $M(n, k_1, k_2, \ldots, k_t)$ reduces to determining $M(n, k_2, \ldots, k_t)$. Therefore, we conclude that $n \geq k_1 + k_t$ is the natural constraint. In this paper, we determine $M(n, k_1, \ldots, k_t)$ and characterize all extremal systems in the full range $n \geq k_1 + k_t$. Before stating our main results, we require the following definition.

Definition 1.3. *Let $k_1 \geq \cdots \geq k_t$ and $n \geq k_1 + k_t$. According to the relationship of n and k_i, we define integers $w, r, \lambda_1, \ldots, \lambda_{w+1}, s_0, \ldots, s_r, \ell_0, \ldots, \ell_r$ as below.*

- *Let $w \in [0, t-2]$ be the smallest integer such that for any $j > i \geq w+1$, we have $n \geq k_i + k_j$.*
- *Let $\lambda_0 = t$, for each $1 \leq i \leq w+1$, let λ_i be the smallest integer such that $n \geq k_i + k_{\lambda_i+1}$.*
- *Let $s_0 = 0, s_1, s_2, \ldots, s_r$ be integers such that $\lambda_1 = \cdots = \lambda_{s_1} > \lambda_{s_1+1} = \cdots = \lambda_{s_1+s_2} > \cdots > \lambda_{s_1+\cdots+s_{r-1}+1} = \cdots = \lambda_{s_1+\cdots+s_r} = \lambda_{w+1}$ and $s_1 + \cdots + s_r = w+1$.*
- *Let $\ell_j = \sum_{i=0}^{j} s_i$ for each $j \in \{0, 1, \ldots r\}$.*

Once $n, k_1, \ldots, k_t$ are fixed, then $w, r, \lambda_i (0 \leq i \leq w+1), s_j, \ell_j (0 \leq j \leq r)$ are also determined. We have the following observation: Let $t \geq 2$, $k_1 \geq k_2 \geq \cdots \geq k_t$ and $n \geq k_1 + k_t$. Then

$$\max_{i=1}^{w+1}\{k_i + k_{\lambda_i+1}\} \leq n < \min_{i=1}^{w}\{k_i + k_{\lambda_i}\}. \tag{1}$$

In particular, we have $\lambda_w \geq w+1$ since $n < k_w + k_{w+1}$, and $\lambda_{w+1} = w+1$.

Suppose that $r, \ell_0, \ldots, \ell_r, \lambda_{\ell_0}, \ldots, \lambda_{\ell_r}$ are defined as in Definition 1.3. For $j \in [0, r]$, we denote

$$m_j = \sum_{i=1}^{\ell_j}\left(\binom{n}{k_i} - \binom{n-k_t}{k_i}\right) + \sum_{i=\ell_j+1}^{\lambda_{\ell_j}}\binom{n-1}{k_i-1} + \sum_{i=\lambda_{\ell_j}+1}^{t}\binom{n-k_t}{k_i-k_t}. \tag{2}$$

The following main result gives a complete solution for the non-empty pairwise cross intersecting families $\mathcal{F}_1 \subseteq \binom{[n]}{k_1}, \mathcal{F}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{F}_t \subseteq \binom{[n]}{k_t}$ in the full range $n \geq k_1 + k_t$.

Theorem 1.4 (Main result). *Let $t \geq 2$, $k_1 \geq k_2 \geq \cdots \geq k_t$ and $n \geq k_1 + k_t$. Suppose that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an $(n, k_1, \ldots, k_t)$-cross intersecting system, then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \max\Big\{m_0, m_1, \ldots, m_r\Big\}.$$

Before characterizing the extremal systems, we introduce the following systems first.

Construction 1.5. *Let $n \geq k_1 + k_t$ and $k_1 \geq k_2 \geq \cdots \geq k_t$. Suppose that $r, \ell_0, \ldots, \ell_r, \lambda_{\ell_0}, \ldots, \lambda_{\ell_r}$ are defined as in Definition 1.3. Let $j \in [0, r]$. We define $\mathcal{A}_{j,1}, \mathcal{A}_{j,2}, \ldots, \mathcal{A}_{j,t}$ as follows.*

- *For each $i \in [\ell_j]$, let $\mathcal{A}_{j,i} = \left\{A \in \binom{[n]}{k_i} : A \cap [k_t] \neq \emptyset\right\}$.*

- *For each* $i \in [\lambda_{\ell_j}+1, t]$, *let* $\mathcal{A}_{j,i} = \left\{A \in \binom{[n]}{k_i} : [k_t] \subseteq A\right\}$.
- *For each* $i \in [\ell_j+1, \lambda_{\ell_j}]$, *let* $\mathcal{A}_{j,i} = \left\{A \in \binom{[n]}{k_i} : 1 \in A\right\}$.

For each $j \in [0,r]$, we have $\sum_{i=1}^{t} |\mathcal{A}_{j,i}| = m_j$. By (1), it is easy to see that $(\mathcal{A}_{j,1}, \mathcal{A}_{j,2}, \ldots, \mathcal{A}_{j,t})$ is an $(n, k_1, \ldots, k_t)$-cross intersecting system for each $j \in [0,r]$. Theorem 1.4 implies that $(\mathcal{A}_{j,1}, \mathcal{A}_{j,2}, \ldots, \mathcal{A}_{j,t})$ is an extremal system. But Construction 1.5 is not the only extremal system. For a family $\mathcal{A}$, we denote $\overline{\mathcal{A}} := \{[n] \setminus A : A \in \mathcal{A}\}$. With this notation, let us consider a special case: Suppose that for each $i \in [r]$, $s_i = \lambda_{\ell_{i-1}} - \lambda_{\ell_i}$, $k_{\ell_{i-1}+1} = \cdots = k_{\ell_i}$, $k_{\lambda_{\ell_i}+1} = \cdots = k_{\lambda_{\ell_{i-1}}}$ and $n = k_{\ell_i} + k_{\lambda_{\ell_i}+1}$. Now, we define the following cross intersecting system.

Construction 1.6. *Suppose that for each* $i \in [r]$, $s_i = \lambda_{\ell_{i-1}} - \lambda_{\ell_i}$, $k_{\ell_{i-1}+1} = \cdots = k_{\ell_i}$, $k_{\lambda_{\ell_i}+1} = \cdots = k_{\lambda_{\ell_{i-1}}}$, *and* $n = k_{\ell_i} + k_{\lambda_{\ell_i}+1}$. *If* $t = 2$, *then let* $\mathcal{D}_1 \subsetneqq \binom{[n]}{k_1}$ *be a non-empty family, and let* $\mathcal{D}_2 = \binom{[n]}{k_2} \setminus \overline{\mathcal{D}_1}$. *Suppose that* $t > 2$.

- *Let* $\mathcal{D}_1$ *be a family that satisfies:* $\{A : 1 \in A, |A| = k_1\} \subseteq \mathcal{D}_1 \subseteq \{A : A \cap [k_t] \neq \emptyset, |A| = k_1\}$; *let* $\mathcal{D}_{\lambda_1+1} = \binom{[n]}{k_{\lambda_1+1}} \setminus \overline{\mathcal{D}_1}$.
- *For any* $i \in \{\ell_1+1, \ell_2+1, \ldots, \ell_{r-1}+1\}$, *let* $\mathcal{D}'_i$ *be the* k_i*-uniform family that is maximal to* $\mathcal{D}_{\lambda_{i-1}+1}$, *and let* $\mathcal{D}_i$ *satisfy* $\{A : 1 \in A, |A| = k_i\} \subseteq \mathcal{D}_i \subseteq \mathcal{D}'_i$ *and* $\mathcal{D}_{\lambda_i+1} = \binom{[n]}{k_{\lambda_i+1}} \setminus \overline{\mathcal{D}_i}$; *or in addition, if* $s_r = 1$, *then let* $\mathcal{D}'_{w+2}$ *be the* k_{w+2}*-uniform family that is maximal to* $\mathcal{D}_{w+3}$, *let* $\mathcal{D}_{w+2}$ *satisfies* $\{A : 1 \in A, |A| = k_{w+2}\} \subseteq \mathcal{D}_{w+2} \subseteq \mathcal{D}'_{w+2}$ *and* $\mathcal{D}_{w+1} = \binom{[n]}{k_{w+1}} \setminus \overline{\mathcal{D}_{w+2}}$.
- *For any* $i \in [r]$, *let* $\mathcal{D}_j = \mathcal{D}_{\ell_{i-1}+1}$ *for any* $j \in [\ell_{i-1}+1, \ell_i]$, *and let* $\mathcal{D}_j = \mathcal{D}_{\lambda_{\ell_{i-1}}+1}$ *for any* $j \in [\lambda_{\ell_i}+1, \ell_{i-1}]$.

Now, we characterize all extremal systems of Theorem 1.4.

Theorem 1.7. *Let* $k_1 \geq \cdots \geq k_t$, $n \geq k_1 + k_t$, *and* $r, \ell_0, \ldots, \ell_r, \lambda_{\ell_0}, \ldots, \lambda_{\ell_r}$ *be defined as in Definition 1.3. Let* $m_u = \max_{0 \leq j \leq r}\{m_j\}$ *for some* $u \in [0,r]$. *Suppose* $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ *is an* $(n, k_1, \ldots, k_t)$*-cross intersecting system with* $\sum_{i=1}^{t} |\mathcal{F}_i| = m_u$. *Then one of the following holds.*

(i) For each $i \in [t]$, $\mathcal{F}_i$ *is isomorphic to* $\mathcal{A}_{u,i}$ *in Construction 1.5.*

(ii) Suppose $u = 0$. *Let* $1 \leq v \leq r$ *be the largest integer such that* $k_1, \ldots, k_{\ell_v}, k_{\lambda_{\ell_v}+1}, \ldots, k_t$ *satisfy the condition of Construction 1.6. Then* $\mathcal{F}_1, \ldots, \mathcal{F}_{\ell_v}, \mathcal{F}_{\lambda_{\ell_v}+1}, \ldots, \mathcal{F}_t$ *are isomorphic to the families in Construction 1.6 (replacing* $k_1, \ldots, k_t$ *in Construction 1.6 by* $k_1, \ldots, k_{\ell_v}$, $k_{\lambda_{\ell_v}+1}, \ldots, k_t$*), and for all* $i \in [\ell_v+1, \lambda_{\ell_v}]$, $\mathcal{F}_i$ *is isomorphic to* $\mathcal{A}_{0,i}$ *in Construction 1.5.*

(iii) Suppose $u = r$. *Let* $0 \leq v \leq r-1$ *be the smallest integer such that* $k_{\ell_v+1}, \ldots, k_{\lambda_{\ell_v}}$ *satisfy the condition of Construction 1.6. Then* $\mathcal{F}_{\ell_v+1}, \ldots, \mathcal{F}_{\lambda_{\ell_v}}$ *are isomorphic to the families in Construction 1.6 (replacing* $k_1, \ldots, k_t$ *in Construction 1.6 by* $k_{\ell_v+1}, \ldots, k_{\lambda_{\ell_v}}$*), and for all* $i \in [\ell_v] \cup [\lambda_{\ell_v}+1, t]$, $\mathcal{F}_i$ *is isomorphic to* $\mathcal{A}_{r,i}$ *in Construction 1.5.*

(iv) Suppose $0 < u < r$. *Let* $v \in [u+1, r]$ *be the largest integer such that* $k_{\ell_u+1}, \ldots, k_{\ell_v}, k_{\lambda_{\ell_v}+1}, \ldots, k_{\lambda_{\ell_u}}$ *satisfy the condition of Construction 1.6. Then* $\mathcal{F}_{\ell_u+1}, \ldots, \mathcal{F}_v, \mathcal{F}_{\lambda_{\ell_v}+1}, \ldots, \mathcal{F}_{\lambda_{\ell_u}}$ *are isomorphic to the families in Construction 1.6 (replacing* $k_1, \ldots, k_t$ *in Construction 1.6 by* $k_{\ell_u+1}, \ldots, \ell_v, k_{\lambda_{\ell_v}+1}, \ldots, k_{\lambda_{\ell_u}}$*), and for all* $i \in [t] \setminus ([\ell_u+1, \ell_v] \cup [\lambda_{\ell_v}+1, \lambda_{\ell_u}])$, $\mathcal{F}_i$ *is isomorphic to* $\mathcal{A}_{u,i}$.

1.1 Our strategy

For two cross-intersecting families $\mathcal{A}$ and $\mathcal{B}$, if the selection of $\mathcal{A}$ influences the selection of $\mathcal{B}$, we say that $\mathcal{B}$ is constrained by $\mathcal{A}$.

According to Definition 1.3, when $n \geq k_1 + k_2$, we have $w = 0$, $r = 1$, $s_1 = \ell_1 = 1$, and $\lambda_1 = 1$. In this case, for any $i \in [t]$, the family $\mathcal{F}_i$ is constrained by the other $t-1$ families; in particular, $\mathcal{F}_1$ is constrained by $\mathcal{F}_2, \ldots, \mathcal{F}_t$.

When $k_1 + k_3 \leq n < k_1 + k_2$, we have $w = 1$, $r = 1$, $s_1 = \ell_1 = 2$, and $\lambda_1 = \lambda_2 = 2$. Here, $\mathcal{F}_1$ and $\mathcal{F}_2$ are not constrained by each other, but both are constrained by $\mathcal{F}_3, \ldots, \mathcal{F}_t$.

The most general case occurs when $n \geq k_1 + k_t$, where r categories arise. More precisely:

$\mathcal{F}_1, \ldots, \mathcal{F}_{\ell_1}$ are constrained by $\mathcal{F}_{\lambda_1+1}, \ldots, \mathcal{F}_t$;

$\mathcal{F}_{\ell_1+1}, \ldots, \mathcal{F}_{\ell_2}$ are constrained by $\mathcal{F}_{\lambda_{\ell_2}+1}, \ldots, \mathcal{F}_t$;

$\cdots$

$\mathcal{F}_{\ell_{r-1}+1}, \ldots, \mathcal{F}_{w+1}$ are constrained by $\mathcal{F}_{w+2}, \ldots, \mathcal{F}_t$.

Since the first $w+1$ families $\mathcal{F}_1, \ldots, \mathcal{F}_{w+1}$ impose no constraints on each other, we can initially establish a multivariate function with the sizes of these $w+1$ families as variables. Our strategy is to reduce this function of $w+1$ variables to a single-variable function by applying extremal constraints. This optimization process differs from the case where $n \geq k_1 + k_3$, as we now face r distinct categories, which presents a significantly more complex challenge. The second challenge lies in determining how this new established single-variable function achieves its maximum. To address this, we formulate Lemma 4.25, which establishes a unimodal-like property for the function. We believe this result itself holds independent interests.

1.2 Organization of the paper

In Sections 2 and 3, we review some foundation work established in [13] and [12]. In Section 4, we will prove Theorems 1.4 and 1.7 by assuming the truth of two crucial lemmas (Lemmas 4.25 and 4.15). In Section 5, we give the proof of Lemma 4.25. In Section 6, we prove Lemma 4.15.

2 Partner and Parity

When we write a set $A = \{a_1, a_2, \ldots, a_s\} \subseteq [n]$, we always assume that $a_1 < a_2 < \ldots < a_s$ throughout the paper. Let $[a, b]$ be the set of integers $\{a, a+1, \ldots, b\}$. By convention, $[a, b] = \emptyset$ when $a > b$. Let $\max A$ denote the maximum element of A, $\min A$ denote the minimum element of A. For two sets A and B, we say that $A \prec B$ in the *lexicographic (lex for short) order* if either $A \supseteq B$ or $\min(A \setminus B) < \min(B \setminus A)$. In particular, $A \prec A$.

Let $\mathcal{L}(n, r, k)$ denote the first r subsets in $\binom{[n]}{k}$ under the lex order. Given a set R, we denote $\mathcal{L}(n, R, k) = \{F \in \binom{[n]}{k} : F \prec R\}$. When the underlying set $[n]$ is clear, we shall write $\mathcal{L}(R, k)$, $\mathcal{L}(r, k)$ for short. Let $\mathcal{F} \subseteq \binom{[n]}{k}$ be a family, we say $\mathcal{F}$ is *L-initial* if $\mathcal{F} = \mathcal{L}(R, k)$ for some k-set R. We call R the **ID** of $\mathcal{F}$.

The well-known Kruskal–Katona theorem [17, 19] allows us to consider only L-initial families when we want to determine $M(n, k_1, \ldots, k_t)$. An equivalent formulation of the Kruskal–Katona theorem was given in [7, 15] as follows.

Theorem 2.1 (Kruskal–Katona theorem [7, 15])**.** *For $\mathcal{A} \subseteq \binom{[n]}{k}$ and $\mathcal{B} \subseteq \binom{[n]}{l}$, if $\mathcal{A}$ and $\mathcal{B}$ are cross intersecting, then $\mathcal{L}(|\mathcal{A}|, k)$ and $\mathcal{L}(|\mathcal{B}|, l)$ are cross intersecting as well.*

Let F and H be two subsets of $[n]$ with size $|F| = f$ and $|H| = h$. We say that F and H *strongly intersect* at their last element if there is an element q such that $F \cap H = \{q\}$ and $F \cup H = [q]$. We also say F is the *partner* of H, or H is the *partner* of F. In [12], we proved the following result.

Proposition 2.2. *[12] Let a, b, n be positive integers and $a + b \le n$. For $P \subseteq [n]$ with $|P| \le a$, let Q be the partner of P. Then $\mathcal{L}(Q, b)$ is the maximum L-initial b-uniform family such that $\mathcal{L}(P, a)$ and $\mathcal{L}(Q, b)$ are cross intersecting.*

In [11], we worked on non-mixed type: Let $t \ge 2$, $k_1 \ge k_2 \ge \cdots \ge k_t$, $n \ge k_1 + k_2$, and let families $\mathcal{A}_1 \subseteq \binom{[n]}{k_1}, \mathcal{A}_2 \subseteq \binom{[n]}{k_2}, \ldots, \mathcal{A}_t \subseteq \binom{[n]}{k_t}$ be non-empty pairwise cross intersecting. The Kruskal–Katona theorem reduces the problem of determining $M(n, k_1, \ldots, k_t)$ to the case where all $\mathcal{A}_i$ are L-initial. Let R be the ID of $\mathcal{A}_1$, and T be the partner of R. To prove Theorem 1.1, one important ingredient is that by Proposition 2.2, $\sum_{i=1}^{t} |\mathcal{A}_i|$ can be bounded by a function of R as following.

$$f(R) = \sum_{j=1}^{t} |\mathcal{A}_j| \le |\mathcal{L}(R, k_1)| + \sum_{j=2}^{t} |\mathcal{L}(T, k_j)|. \tag{3}$$

To prove Theorem 1.4, we may also assume that all $\mathcal{F}_i$ are L-initial. However, when $n \ge k_1 + k_t$, it is not straightforward to obtain a single value function analogous to (3) to bound $\sum_{i=1}^{t} |\mathcal{F}_i|$. To address this issue, we established some properties in [13], they are parts of the foundation of our main results. In this section, we review these properties. Let's introduce some concepts first.

- **$\mathcal{F}$ is maximal to $\mathcal{G}$.** Let $\mathcal{F} \subseteq \binom{[n]}{f}$ and $\mathcal{G} \subseteq \binom{[n]}{g}$ be cross intersecting families. We say that $\mathcal{F}$ *is maximal to* $\mathcal{G}$ if whenever $\mathcal{F}' \subseteq \binom{[n]}{f}$ and $\mathcal{G}$ are cross intersecting with $\mathcal{F} \subseteq \mathcal{F}'$, then $\mathcal{F}' = \mathcal{F}$.

- **$(\mathcal{F}, \mathcal{G})$ is maximal.** $(\mathcal{F}, \mathcal{G})$ is maximal if and only if $\mathcal{F}$ is maximal to $\mathcal{G}$ and $\mathcal{G}$ is maximal to $\mathcal{F}$.

- **F is maximal to G.** Let F and G be two subsets of $[n]$. We say that F *is maximal to* G if there are two L-initial families $\mathcal{F} \subseteq \binom{[n]}{|F|}$ and $\mathcal{G} \subseteq \binom{[n]}{|G|}$ with IDs F and G respectively such that $\mathcal{F}$ is maximal to $\mathcal{G}$.

- **(F, G) is maximal.** (F, G) is maximal if and only if F is maximal to G and G is maximal to F.

- **maximal pair families.** We say two families $\mathcal{A}_1$ and $\mathcal{A}_2$ are *maximal pair families* if $|\mathcal{A}_1| = |\mathcal{A}_2|$ and for every $A_1 \in \mathcal{A}_1$, there is a unique $A_2 \in \mathcal{A}_2$ such that (A_1, A_2) is maximal.

- Let $F = \{x_1, x_2, \ldots, x_k\} \subseteq [n]$. We denote

$$\ell(F) = \begin{cases} \max\{x : [n - x + 1, n] \subseteq F\}, & \text{if } \max F = n, \\ 0, & \text{if } \max F < n, \end{cases}$$
$$F^{\mathrm{t}} = [n - \ell(F) + 1, n].$$

- **k-partner.** Let F and H be two subsets of $[n]$ and partners to each other with $|F| = f$ and $|H| = h$. Let $k \le n - f$ be an integer, we define the *k-partner* K of F as follows. For $k = h$, let $K = H$. If $k > h$, then let $K = H \cup \{n - k + h + 1, \ldots, n\}$. If $k < h$, then let K be the last k-set in $\binom{[n]}{k}$ such that $K \prec H$, in other words, there is no k-set K' satisfying $K \precneqq K' \precneqq H$. We claim that $|K| = k$. To see this, we only need to check the case $k > h$. Indeed, since F and H intersect at their last element, $n' = \max H = f + h - 1 < n - k + h$. This implies $K^{\mathrm{t}} = \{n - k + h + 1, \ldots, n\}$ and $|K| = |H \cup \{n - k + h + 1, \ldots, n\}| = k$.

- **parity.** Let h_1 and h_2 be positive integers with $h_1 \le h_2$, H_1 and H_2 be subsets of $[n]$ with sizes h_1 and h_2, respectively. We say H_1 is the *h_1-parity* of H_2 (or H_2 is the h_2-parity of H_1) if $H_1 \setminus H_1^{\mathrm{t}} = H_2 \setminus H_2^{\mathrm{t}}$ and $\ell(H_2) - \ell(H_1) = h_2 - h_1$.

Fact 2.3. *[13] Let a, b, n be positive integers and $n \geq a + b$. For $A \subseteq [n]$ with $|A| = a$, let B be the b-partner of A, then $\mathcal{L}(B, b)$ is maximal to $\mathcal{L}(A, a)$, i.e., B is maximal to A.*

Note that families $\mathcal{L}(A, a)$ and $\mathcal{L}(B, b)$ which mentioned in Fact 2.3 may not be maximal cross intersecting. Although $\mathcal{L}(B, b)$ is maximal to $\mathcal{L}(A, a)$, $\mathcal{L}(A, a)$ may not be maximal to $\mathcal{L}(B, b)$. For example, let $n = 9$, $a = 3$, $b = 4$ and $A = \{2, 4, 7\}$. Then the b-partner of A is $\{1, 3, 4, 9\}$. Although $\mathcal{L}(\{1, 3, 4, 9\}, 4)$ is maximal to $\mathcal{L}(\{2, 4, 7\}, 3)$, $\mathcal{L}(\{1, 3, 4, 9\}, 4)$ and $\mathcal{L}(\{2, 4, 7\}, 3)$ are not maximal cross intersecting families since $\mathcal{L}(\{2, 4, 7\}, 3) \subsetneqq \mathcal{L}(\{2, 4, 9\}, 3)$, and $\mathcal{L}(\{2, 4, 9\}, 3)$ and $\mathcal{L}(\{1, 3, 4, 9\}, 4)$ are cross intersecting families.

Frankl–Kupavskii [6] gave a sufficient condition for a pair of maximal cross intersecting families as stated below.

Proposition 2.4 (Frankl–Kupavskii [6])**.** *Let $a, b \in \mathbb{Z}_{>0}, a + b \leq n$. Let P and Q be non-empty subsets of $[n]$ with $|P| \leq a$ and $|Q| \leq b$. If Q is the partner of P, then $\mathcal{L}(P, a)$ and $\mathcal{L}(Q, b)$ are maximal cross intersecting families.*

Based on Proposition 2.4, we gave a necessary and sufficient condition for a pair of maximal cross intersecting families in terms of their IDs in [13].

Proposition 2.5. *[13] Let $n = k + \ell$, A be any k-subset of $[n]$. Then there is an ℓ-set $B \subseteq [n]$ such that (A, B) is maximal. Moreover, for any $A \in \binom{[n]}{k}$, let B be the ℓ-set such that (A, B) is maximal, then we have $M(n, k, \ell) = |\mathcal{L}(A, k)| + |\mathcal{L}(B, \ell)| = \binom{n}{\ell}$.*

Fact 2.6. *[13] Let a, b, n be positive integers and $n \geq a+b$. Suppose that A is an a-subset of $[n]$, and B is the b-partner of A. Let A' be the a-partner of B, then (A', B) is maximal. Moreover, if $A' \neq A$, then $A \supsetneqq A'$.*

Fact 2.7. *[13] Let a, b, k, n be positive integers and $n \geq \max\{a + k, b + k\}$. Let A and B be two subsets of $[n]$ with sizes $|A| = a$ and $|B| = b$. Suppose that K_a and K_b are the k-partners of A and B respectively. If $A \prec B$, then $K_b \prec K_a$. In particular, if A is the a-parity of B, then $K_b = K_a$.*

Fact 2.8. *[13] Let a, b, c, n be positive integers, $a \geq b$ and $n \geq a + c$, and let C be a c-subset of $[n]$. Suppose that A is the a-partner of C and B is the b-partner of C, then $B \prec A$ or A is the a-parity of B.*

Definition 2.9. *Let $h_1 \leq h_2$. For two families $\mathcal{H}_1 \subseteq \binom{[n]}{h_1}$ and $\mathcal{H}_2 \subseteq \binom{[n]}{h_2}$, we say that $\mathcal{H}_1$ is the h_1-parity of $\mathcal{H}_2$, or $\mathcal{H}_2$ is the h_2-parity of $\mathcal{H}_1$ if the following properties hold.*

(i) For any $H_1 \in \mathcal{H}_1$, the h_2-parity of H_1 exists and must be in $\mathcal{H}_2$.

(ii) For any $H_2 \in \mathcal{H}_2$, either H_2 has no h_1-parity or its h_1-parity is in $\mathcal{H}_1$.

Proposition 2.10. *[13] Let f, g, h, n be positive integers with $f \geq g$ and $n \geq f + h$. Let*

$$\mathcal{F} = \left\{ F \in \binom{[n]}{f} : \text{ there exists } H \in \binom{[n]}{h} \text{ such that } (F, H) \text{ is maximal} \right\},$$
$$\mathcal{G} = \left\{ G \in \binom{[n]}{g} : \text{ there exists } H \in \binom{[n]}{h} \text{ such that } (G, H) \text{ is maximal} \right\}.$$

Let $\mathcal{H}_{\mathcal{F}} \subseteq \binom{[n]}{h}$ and $\mathcal{H}_{\mathcal{G}} \subseteq \binom{[n]}{h}$ be the families such that $\mathcal{F}$ and $\mathcal{H}_{\mathcal{F}}$ are maximal pair families and $\mathcal{G}$ and $\mathcal{H}_{\mathcal{G}}$ are maximal pair families. Then $\mathcal{F}$ is the f-parity of $\mathcal{G}$; $\mathcal{H}_{\mathcal{G}} \subseteq \mathcal{H}_{\mathcal{F}}$; and for any $G \in \mathcal{G}$, let $F \in \mathcal{F}$ be the f-parity of G and $H \in \mathcal{H}$ such that (F, H) is maximal, then (G, H) is maximal.

3 Preliminary

In this section, we summarize some key results from our earlier work on the non-mixed type [12] and on the first mixed type [13], these results will be utilized in the present paper.

Definition 3.1 (c-**sequential**). *Let $\mathcal{A} \subseteq \binom{[n]}{k}$ and $c \in [k]$. We say that $\mathcal{A}$ is c-sequential if there are $A \subseteq [n]$ with $|A| = k - c$ and $a \geq \max A$ such that $\mathcal{A} = \{A \sqcup \{a+1, \ldots, a+c\}, A \sqcup \{a+2, \ldots, a+c+1\}, \ldots, A \sqcup \{b-c+1, \ldots, b\}\}$. We write $A_1 \overset{c}{\prec} A_2 \overset{c}{\prec} \cdots \overset{c}{\prec} A_{b-a-c+1}$, where $A_1 = A \sqcup \{a+1, \ldots, a+c\}$, $A_2 = A \sqcup \{a+2, \ldots, a+c+1\}, \ldots, A_{b-a-c+1} = A \sqcup \{b-c+1, \ldots, b\}$.*

Definition 3.2 ($F_1 < F_2$). *We say that $F_1 < F_2$ if $F_1 \precneqq F_2$ and there is no F' such that $|F'| = |F_1|$ and $F_1 \precneqq F' \precneqq F_2$.*

3.1 Some results in [12]

Let $t, n, k_1, k_2, \ldots, k_t$ be positive integers with $k_1 \geq k_2 \geq \cdots \geq k_t$, $t \geq 2$ and $n \geq k_1 + k_2$. Denote

$$\mathbb{R}_1 = \left\{ R \in \binom{n}{k_1} : \{1, n-k_1+2, \ldots, n\} \prec R \prec \{k_t, n-k_1+2, \ldots, n\} \right\}.$$

Let $t \geq 2$, $k_1, k_2, \ldots, k_t$ be positive integers with $k_1 \geq k_2 \geq \cdots \geq k_t$ and $n \geq k_1 + k_2$. Let R and R' be of $\mathbb{R}_1$ such that $R \prec R'$. For each $j \in [2, t]$, let T_j and T'_j be the k_j-partners of R and R', respectively. We define*

$$\alpha(R, R') = |\mathcal{L}(R', k_1)| - |\mathcal{L}(R, k_1)|, \tag{4}$$

$$\beta(R, R') = \sum_{j=2}^{t} \big(|\mathcal{L}(T_j, k_j)| - |\mathcal{L}(T'_j, k_j)| \big), \tag{5}$$

$$f_1(R) = |\mathcal{L}(R, k_1)| + \sum_{j \neq i} |\mathcal{L}(T_j, k_j)|. \tag{6}$$

The authors proved the following result which constitutes an essential part of the 'unimodality method' in [12]. (In fact, [12] presents a more general form.)

Proposition 3.3 (Proposition 2.12 in [12]). *Let $n \geq k_1 + k_2$ and $k_1 \geq k_2 \geq \cdots \geq k_t$. Let $F, G \in \mathbb{R}_1$, $F < G$, and $\max G = q$. Then*
(i) $\alpha(F, G) = 1$.
(ii) $\beta(F, G) = \sum_{j=2}^{t} \binom{n-q}{k_j - (q-k_1)}$.
(iii) If $n = k_1 + k_j$ holds for any $j \in [t] \setminus \{1\}$, then $\beta(F, G) = t - 1$; otherwise, $\beta(F, G)$ decreases when q strictly increases until $\beta(F, G) = 0$.

Proposition 3.4 (Proposition 3.1 in [12]). *Suppose that $n \geq k_1 + k_2$, $k_1 \geq k_2 \geq \cdots \geq k_t$ are positive integers. Let $m_i = \min\{k_j : j \in [t] \setminus \{i\}\}$. For each $x \in [m_i]$ and each $i \in [t]$, let*

$$f_i(\{x\}) = \binom{n}{k_i} - \binom{n-x}{k_i} + \sum_{j \neq i} \binom{n-x}{k_j - x}.$$

Then

$$f_1(\{x\}) = \max\{f_j(\{s\}) : j \in [t]\} \text{ for each } x \in [k_t],$$
$$f_t(\{k_{t-1}\}) \leq \max\{f_1(\{1\}), f_1(\{k_t\})\}.$$

*Taking $i = 1$, $d_1 = \cdots = d_t = 1$ and $m_i = k_t$ in reference [12, eq. (16) and (17)].

3.2 Some results in [13]

Definition 3.5. *Suppose that $t \geq 2$, $k_1 \geq k_2 \geq \cdots \geq k_t$, $s \in [t-1]$ and $k_1+k_{s+1} \leq n < k_{s-1}+k_s$. We define $\mathcal{R}_i$ for every $i \in [t]$ as follows. For $i \in [s]$, let*

$$\mathcal{R}_i = \left\{ R \in \binom{[n]}{k_i} : \{1, n-k_i+2, \ldots, n\} \prec R \prec \{k_t, n-k_i+2, \ldots, n\} \right\}.$$

For $i \in [s+1, t]$, let

$$\mathcal{R}_i = \left\{ R \in \binom{[n]}{k_i} : [k_t] \subseteq R \right\}.$$

Definition 3.6. *Let k and n be positive integers. Let $R \subseteq [n]$ be a set. We define the corresponding k-set of R as follows. If R has a k-parity, then let T be the k-parity of R; otherwise, let T be the k-set such that $T < R$. We call T the corresponding k-set of R.*

Definition 3.7. *Let $t \geq 2$, $k_1 \geq k_2 \geq \cdots \geq k_t$, $s \in [t-1]$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let R_1, $R_1' \in \mathcal{R}_1$ with $R_1 \prec R_1'$ and for any $i \in [s+1, t]$, R_i, R_i' be the k_i-partner of R_1, R_1' respectively, and for each $i \in [2, s]$, let the corresponding R_i, R_i' be defined as Definition 3.6. We define the following functions.*

$$\alpha_i(R_1, R_1') = |\mathcal{L}(R_i', k_i)| - |\mathcal{L}(R_i, k_i)| \text{ for each } i \in [s],$$

$$\gamma(R_1, R_1') = \sum_{i=1}^{s} \alpha_i(R_1, R_1'),$$

$$\delta(R_1, R_1') = \sum_{i=s+1}^{t} \big(|\mathcal{L}(R_i, k_i)| - |\mathcal{L}(R_i', k_i)|\big).$$

Remark 3.8. *[13] Let s, s', t be positive integers with $s > s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$ and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let $A_1, B_1, C_1 \in \mathcal{R}_1$ with $A_1 \prec B_1 \prec C_1$. Then for $i \in [s]$, $\alpha_i(A_1, C_1) = \alpha_i(A_1, B_1) + \alpha_i(B_1, C_1)$, $\gamma(A_1, C_1) = \gamma(A_1, B_1) + \gamma(B_1, C_1)$, and $\delta(A_1, C_1) = \delta(A_1, B_1) + \delta(B_1, C_1)$.*

Lemma 3.9. *[13] Let s, s', t be integers with $s \geq s' \geq 1$, $t \geq s+1$, $k_1 = \cdots = k_{s'} > k_{s'+1} \geq \cdots \geq k_t$, and $k_1 + k_{s+1} \leq n < k_{s-1} + k_s$. Let $A_1, B_1, C_1 \in \mathcal{R}_1$ with $A_1 \setminus A_1^{\mathrm{t}} = A \sqcup \{a, a+1\}$, $B_1 = A \sqcup \{a\} \sqcup [n-\ell(B_1)+1, n]$ and $C_1 = A \sqcup \{a+1\} \sqcup [n-\ell(B_1)+1, n]$ (where $\max A < a$). Then $\delta(A_1, B_1) = \delta(B_1, C_1)$ and $\gamma(A_1, B_1) \leq \gamma(B_1, C_1)$.*

4 Proofs of Theorems 1.4 and 1.7

Let integers $n, t, w, r, k_i, \lambda_i (i \in [t])$, $s_i (i \in [0, r])$ be the constants in Theorem 1.4, and let $m_0, \ldots, m_r$ be integers defined in (2).

If $w = 0$ or $t \leq 3$, then by Theorems 1.1 and 1.2, we are done. Thus we may assume that $t \geq 4$ and $w \geq 1$.

We first consider the case: $k_t = 1$. Suppose $|\mathcal{F}_t| = c$. Then $\mathcal{F}_t = \{\{1\}, \ldots, \{c\}\}$. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an extremal non-empty $(n, k_1, \ldots, k_t)$-cross intersecting system, then for any $i \in [t-1]$ and any $F \in \mathcal{F}_i$, we have $[c] \subseteq F$ and $1 \leq c \leq k_{t-1}$. Thus,

$$\sum_{i=1}^{t} |\mathcal{F}_i| = \sum_{i=1}^{t-1} \binom{n-c}{k_i - c} + c \leq m_0,$$

as desired. Note that the above equality holds if and only if $c = 1$, that is $\mathcal{F}_t = \{x\}$ for some $x \in [n]$. Therefore, $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to $(\mathcal{A}_{0,1}, \ldots, \mathcal{A}_{0,t})$, which is defined in Construction 1.5, we are done. From now on, we may assume that

$$k_t \geq 2,\ t \geq 4 \text{ and } w \geq 1.$$

We need the following result which is a generalization of the Erdős–Ko–Rado theorem.

Theorem 4.1 (Daykin [4]). *Let $n \geq a+b$, $\mathcal{A} \subseteq \binom{[n]}{a}$ and $\mathcal{B} \subseteq \binom{[n]}{b}$ be crossing intersecting. If $|\mathcal{A}| \geq \binom{n-1}{a-1}$ (or $|\mathcal{A}| > \binom{n-1}{a-1}$), then $|\mathcal{B}| \leq \binom{n-1}{b-1}$ ($|\mathcal{B}| < \binom{n-1}{b-1}$ respectively).*

Let us consider a special case first.

Proposition 4.2. *Under the conditions in Theorem 1.4, suppose further that for each $i \in [r]$, $s_i = \lambda_{\ell_{i-1}} - \lambda_{\ell_i}$, $k_{\ell_{i-1}+1} = \cdots = k_{\ell_i}$, $k_{\lambda_{\ell_i}+1} = \cdots = k_{\lambda_{\ell_{i-1}}}$, and $n = k_{\ell_i} + k_{\lambda_{\ell_i}+1}$. Then*

$$\sum_{i=1}^{t} |\mathcal{F}_i| \leq \sum_{i=1}^{t} \binom{n-1}{k_i - 1},$$

and equality holds if and only if $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to the system that defined in Construction 1.6.

Proof. In this case, $m_0 = m_1 = \cdots = m_r$. Since $s_i = \lambda_{\ell_{i-1}} - \lambda_{\ell_i}$ for each $i \in [r]$, $t = 2(w+1)$ is even. It it obvious if $t = 2$. So we may assume $t \geq 4$. From Proposition 2.5, we obtain that for each $i \in [w+1]$, $|\mathcal{F}_i| + |\mathcal{F}_{\lambda_i+1}| \leq \binom{n}{k_i}$. Therefore, $\sum_{i=1}^{t} |\mathcal{F}_i| \leq \sum_{i=1}^{w+1} \binom{n}{k_i}$, equality holds if and only if $|\mathcal{F}_i| + |\mathcal{F}_{\lambda_i+1}| = \binom{n}{k_i}$ holds for each $i \in [w+1]$. Thus, if $s_r > 1$, then for each $i \in [w+2, t]$ (if $s_r = 1$, then for each $i \in [w+3, t]$), $\mathcal{F}_i \subseteq \{F : x \in F, |F| = k_i\}$ for some x. Indeed, if there is $i \in [w+2, t]$ (for the case $s_r = 1$, here is $i \in [w+3, t]$) such that $|\mathcal{F}_i| > \binom{n-1}{k_i-1}$, then by Theorem 4.1, there is $j \in [w+1]$ such that $|\mathcal{F}_j| < \binom{n-1}{k_j-1}$ and $|\mathcal{F}_{\lambda_j+1}| < \binom{n-1}{k_{\lambda_j+1}-1}$, a contradiction. This implies that either for all $i \in [w+1]$ (if $s_r > 1$) or for all $i \in [w] \cup \{w+2\}$ (if $s_r = 1$), $\{F : x \in F, |F| = k_i\} \subseteq \mathcal{F}_i$. Since $n = k_1 + k_t = k_2 + k_{t-1} = \cdots = k_{w+1} + k_{w+2}$, by Proposition 2.5 again, we will get that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to $(\mathcal{D}_1, \ldots, \mathcal{D}_t)$ in Construction 1.6. □

We will first show the conclusions of Theorems 1.4 and 1.7 hold for the following so called '$(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t)$-*cross intersecting system*' under some conditions.

Definition 4.3. *Suppose that $\mathcal{A}_1 \subseteq \binom{[n]}{k_1}, \ldots, \mathcal{A}_t \subseteq \binom{[n]}{k_t}$ are non-empty pairwise cross intersecting with $k_1 \geq \cdots \geq k_i$ and $k_{i+1} \geq \cdots \geq k_t$, and $|\mathcal{A}_j| \geq \binom{n-1}{k_j-1}$ for each $j \in [i]$, then we call $(\mathcal{A}_1, \ldots, \mathcal{A}_t)$ an $(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t)$-cross intersecting system. Denote*

$$M(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t) = \max \Bigg\{ \sum_{i=1}^{t} |\mathcal{A}_i| : (\mathcal{A}_1, \ldots, \mathcal{A}_t) \text{ is an } (n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t)\text{-crossintersecting system.} \Bigg\}.$$

We say that an $(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t)$-cross intersecting system $(\mathcal{A}_1, \ldots, \mathcal{A}_t)$ is extremal *if $\sum_{i=1}^{t} |\mathcal{A}_i| = M(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t)$.*

Assumption 4.4. *Let integers $0 \leq s \leq t-2$, $n \geq k_1 + k_t$, $k_1 \geq \cdots \geq k_{s+1}$ and $k_{s+2} \geq \cdots \geq k_t \geq 2$. Let $\lambda_0 = t$, for each $1 \leq i \leq s+1$, let λ_i be the smallest integer in $[s+1, t]$ such that $n \geq k_i + k_{\lambda_i+1}$. Let $s_0 = 0$, $s_1, s_2, \ldots, s_r$ be the integers such that $\lambda_1 = \cdots = \lambda_{s_1} > \lambda_{s_1+1} = \cdots = \lambda_{s_1+s_2} > \cdots > \lambda_{s_1+\cdots+s_{r-1}+1} = \cdots = \lambda_{s+1} = s+1$ and $s_0 + s_1 + \cdots + s_r = s+1$. Let $\ell_j = \sum_{i=0}^{j} s_i$ for each $j \in \{0, 1, \ldots r\}$.*

Remark 4.5. *In the above assumption, we require the weaker conditions $k_1 \geq \cdots \geq k_{s+1}$ and $k_{s+2} \geq \cdots \geq k_t$ than $k_1 \geq \cdots \geq k_t$, we also have $\max_{i=1}^{s+1}\{k_i + k_{\lambda_i+1}\} \leq n < \min_{i=1}^{s}\{k_i + k_{\lambda_i}\}$, which is analogous to (1). In particular, we have $\lambda_s \geq s+1$ and $n < k_s + k_{s+1}$.*

The following proposition will be frequently used in our proof.

Proposition 4.6. *Under Assumption 4.4, suppose that $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is an extremal L-initial $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system with ID F_i of $\mathcal{F}_i$ for each $i\in[t]$. Fix an arbitrary p in $[s+1]$, and let x be the integer such that $p\in[\ell_{x-1}+1,\ell_x]$. Then the following properties hold.*

(i) If $F_p=\{k_t,n-k_p+2,\dots,n\}$, then for each $j\in[\ell_x]$, $F_j=\{k_t,n-k_j+2,\dots,n\}$; for each $j\in[\lambda_{\ell_x}+1,t]$, $F_j=\{1,\dots,k_t,n-k_j+k_t+1,\dots,n\}$.

(ii) If $F_p=\{1,n-k_p+2,\dots,n\}$, then for each $j\in[\ell_{x-1}+1,\lambda_{\ell_{x-1}}]$, $F_j=\{1,n-k_j+2,\dots,n\}$, and $(\mathcal{F}_1,\dots,\mathcal{F}_{\ell_{x-1}},\mathcal{F}_{\lambda_{\ell_{x-1}}+1},\mathcal{F}_t)$ is an extremal L-initial $(n,k_1,\dots,k_{\ell_{x-1}}||k_{\lambda_{\ell_{x-1}}+1},\dots,k_t)$-cross intersecting system.

Proof. Since $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is an extremal L-initial $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system, we have

$$\{1,n-k_i+2,\dots,n\}\prec F_i\prec\{k_t,n-k_i+2,\dots,n\}\text{ for each }i\in[s+1],$$
$$\{1,\dots,k_t,n-k_j+k_t+1,\dots,n\}\prec F_j\prec\{1,n-k_j+2,\dots,n\}\text{ for each }i\in[s+2,t]. \tag{7}$$

Suppose $F_p=\{k_t,n-k_p+2,\dots,n\}$ first. For each $j\in[\lambda_p+1,t]$, since $\mathcal{F}_j$ and $\mathcal{F}_p$ are cross intersecting and $n\geq k_p+k_{\lambda_p+1}\geq k_p+k_j$, every member of $\mathcal{F}_j$ must contain subset $\{1,\dots,k_t\}$. Since (7) and $\lambda_p=\lambda_{\ell_x}$, we get

$$F_j=\{1,\dots,k_t,n-k_j+k_t+1,\dots,n\}\text{ for each }j\in[\lambda_{\ell_x}+1,t], \tag{8}$$

as required.

Note that for all $j\in[\ell_x]$, we have $\lambda_j\geq\lambda_p$, $n\geq k_j+k_{\lambda_j+1}$, and $\mathcal{F}_j$ is cross intersecting with $\mathcal{F}_i$ for each $i\in[\lambda_j+1,t]$. Since $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is extremal and (8), $F_j=\{k_t,n-k_j+2,\dots,n\}$ for each $j\in[\ell_x]$, as required.

Next, we consider the case: $F_p=\{1,n-k_p+2,\dots,n\}$. Suppose on the contrary that there is $j\in[\ell_{x-1}+1,s+1]$, such that $\{1,n-k_j+2,\dots,n\}\precneqq F_j$. We may choose F_j such that $F_{j'}\prec F_j$ for all $j'\in[\ell_{x-1}+1,s+1]$. Combining Facts 2.7, 2.3 and (1) , since $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is extremal, we conclude that for each $k\in[\lambda_{\ell_x}+1,\lambda_{\ell_{x-1}}]$, F_k is the k_k-partner of F_j, and $F_k\prec F_{\lambda_{\ell_x}+1}\precneqq\{1,n-k_{\lambda_{\ell_x}+1}+2,\dots,n\}$. For each $k\in[\lambda_{\ell_x}+1,\lambda_{\ell_{x-1}}]$, let $F_{p,k}$ be the k_p-partner of F_k. Then Fact 2.8 gives $\{1,n-k_p+2,\dots,n\}\precneqq F_{p,\lambda_{\ell_x}+1}\prec F_{p,k}$ for all $k\in[\lambda_{\ell_x}+1,\lambda_{\ell_{x-1}}]$. This implies $F_p\precneqq F_{p,\lambda_{\ell_x}+1}$, and if we replace F_p by $F_{p,\lambda_{\ell_x}+1}$, then the new system is still cross intersecting. This is a contradiction to the fact that $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is extremal. So for each $k\in[\ell_{x-1}+1,s+1]$, $F_k=\{1,n-k_k+2,\dots,n\}$. By (1) , we get that for each $k\in[s+2,\lambda_{\ell_{x-1}}]$, $F_k=\{1,n-k_k+2,\dots,n\}$ as well. As a consequence, $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is extremal implies that $(\mathcal{F}_1,\dots,\mathcal{F}_{\ell_{x-1}},\mathcal{F}_{\lambda_{\ell_{x-1}}+1},\mathcal{F}_t)$ is an extremal L-initial $(n,k_1,\dots,k_{\ell_{x-1}}||k_{\lambda_{\ell_{x-1}}+1},\dots,k_t)$-cross intersecting system, we are done. □

Assumption 4.7. *Under Assumption 4.4, suppose further that for each $i\in[r]$, if $s_i=\lambda_{\ell_{i-1}}-\lambda_{\ell_i}$, $k_{\ell_{i-1}+1}=\cdots=k_{\ell_i}$ and $k_{\lambda_{\ell_i}+1}=\cdots=k_{\lambda_{\ell_{i-1}}}$, then $n>k_{\ell_i}+k_{\lambda_{\ell_i}+1}$.*

Remark 4.8. *Under Assumptions 4.4 and 4.7, if $n=k_{\ell_i}+k_{\lambda_{\ell_i}+1}$ for some $i\in[r]$, then $k_{\ell_{i-1}+1}=\cdots=k_{\ell_i}$ and $k_{\lambda_{\ell_i}+1}=\cdots=k_{\lambda_{\ell_{i-1}}}$, therefore, Assumption 4.7 implies that $s_i\neq\lambda_{\ell_{i-1}}-\lambda_{\ell_i}$ in this case.*

We are going to prove the following theorem.

Theorem 4.9. *Suppose that $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is an extremal $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system satisfying Assumptions 4.4 and 4.7. Then*

$$\sum_{i=1}^{t}|\mathcal{F}_i|=\max\{m_0,m_1,\dots,m_r\}.$$

Equality holds if and only if $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is isomorphic to the system given in Construction 1.5.

4.1 The proof of the quantitative part of Theorem 4.9

In this section, we always assume that $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is an extremal $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system satisfying Assumptions 4.4 and 4.7. By the Kruskal–Katona theorem, to prove the quantitative part of Theorem 4.9, we may assume that $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ **is L-initial with *ID* F_j for each** $j\in[t]$. Let us look at where these IDs lie. For $i\in[s+1]$, we denote

$$\mathbb{R}_i=\left\{R\in\binom{n}{k_i}:\{1,n-k_i+2,\dots,n\}\prec R\prec\{k_t,n-k_i+2,\dots,n\}\right\}.$$

For $i\in[s+2,t]$, denote

$$\mathbb{R}_i=\left\{R\in\binom{n}{k_i}:\{1,\dots,k_t,n-k_i+k_t+1,\dots,n\}\prec R\prec\{1,n-k_i+2,\dots,n\}\right\}.$$

We claim that for each $i\in[t]$, the ID F_i must be in $\mathbb{R}_i$. Let's explain this. Fix an arbitrary $i\in[t]$. Assume that $i\in[s+1]$ first. From Definition 4.3, we have $\{1,n-k_i+2,\dots,n\}\prec F_i$. Since $[k_t]\in\mathcal{F}_t$ and $\mathcal{F}_i$ and $\mathcal{F}_t$ are non-freely cross intersecting, $F_i\prec\{k_t,n-k_i+2,\dots,n\}$. Thus $F_i\in\mathbb{R}_i$, as required. Next, assume that $i\in[s+2,t]$. Since $|\mathcal{F}_j|\geq\binom{n-1}{k_j-1}$ for all $j\in[s+1]$ and $\max_{i=1}^{s+1}\{k_i+k_{\lambda_i+1}\}\leq n<\min_{i=1}^{s}\{k_i+k_{\lambda_i}\}$, then every member in $\mathcal{F}_i$ must contain 1 (in view of Theorem 4.1). On the other hand, for every $j\in[s+1]$, $F_j\prec\{k_t,n-k_i+2,\dots,n\}$ implies $\{F\in\binom{[n]}{k_i}:[k_t]\subseteq F\}\subseteq\mathcal{F}_i$, so $F_i\in\mathbb{R}_i$, as required.

Let $S=\{x_1,\dots,x_d\}\subseteq[t]$ and $\mathcal{A}_x\subseteq\binom{[n]}{k_x}$ for each $x\in S$. If these $\mathcal{A}_x$ are L-initial pairwise cross intersecting with IDs A_x for all $x\in S$, then we denote

$$m(n,A_{x_1},\dots,A_{x_d})=\max\Bigg\{\sum_{j\in[t]\setminus S}|\mathcal{A}_j|:\mathcal{A}_j\subseteq\binom{[n]}{k_j}\text{ is L-initial and }(\mathcal{A}_1,\dots,\mathcal{A}_t)$$
$$\text{is an }(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)\text{-cross intersecting system.}\Bigg\}$$

Note that $n<k_s+k_{s+1}$ and families $\mathcal{F}_1$, $\mathcal{F}_2$, …, $\mathcal{F}_{s+1}$ are pairwise cross intersecting (freely). Therefore, for any given $G_1\in\mathbb{R}_1$, …, $G_{s+1}\in\mathbb{R}_{s+1}$, $s+1$ L-initial families with IDs G_1, …, G_{s+1}, respectively, are cross intersecting. Denote

$$f(G_1,\dots,G_{s+1})=\sum_{i=1}^{s+1}|\mathcal{L}(G_i,k_i)|+m(n,G_1,\dots,G_{s+1}).$$

Since $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is extremal, we have

$$\sum_{i=1}^{t}|\mathcal{F}_i|=\max\{f(G_1,\dots,G_{s+1}):G_i\in\mathbb{R}_i,i\in[s+1]\}.\tag{9}$$

Proposition 4.10. *Under Assumptions 4.4 and 4.7, additionally, suppose that $k_{\lambda_1+1}=\cdots=k_t$ and $n=k_1+k_{\lambda_1+1}$. Let $d:=|\{i\in[\ell_1]:k_i=k_1\}|$. Let $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ be an L-initial extremal $(n,k_1,\dots,k_t)$-cross intersecting system with ID F_i for each $i\in[t]$. If $d\geq t-\lambda_1$, then for each $i\in[\ell_1]$, $F_i=\{k_t,n-k_i+2,\dots,n\}$.*

Proof. Considering an extremal L-initial $(n,k_1,\dots,k_{\ell_1}||k_{\lambda_1+1},\dots,k_t)$-cross intersecting system $(\mathcal{G}_1,\dots,\mathcal{G}_{\ell_1},\mathcal{G}_{\lambda_1+1},\dots,\mathcal{G}_t)$ with IDs $G_1,\dots,G_{\ell_1},G_{\lambda_1+1},\dots,G_t$ that satisfies the conditions of Proposition 4.10. We have the following claim.

Claim 4.11. *For each $i\in[\ell_1]$, we have $G_i=\{k_t,n-k_i+2,\dots,n\}$.*

Proof. Since $k_{\lambda_1+1} = \cdots = k_t$ and $n = k_1 + k_{\lambda_1+1}$, $|\mathcal{G}_1| + |\mathcal{G}_t| \le \binom{n}{k_1} - \binom{n-k_t}{k_1} + 1$. Thus

$$\sum_{i=1}^{t-\lambda_1} |\mathcal{G}_i| + \sum_{i=\lambda_1+1}^{t} |\mathcal{G}_i| \le (t-\lambda_1)\left(\binom{n}{k_1} - \binom{n-k_t}{k_1}\right) + (t-\lambda_1).$$

Notet that $d \ge t - \lambda_1$ and if $d = t - \lambda_1$, then $k_{\ell_1} < k_1$. For each $i \in [t-\lambda_1, \ell_1]$, we have

$$|\mathcal{G}_i \le \binom{n}{k_i} - \binom{n-k_t}{k_i}.$$

So

$$\sum_{i\in[\ell_1]\cup[\lambda_1+1,t]} |\mathcal{G}_i| \le \sum_{i=1}^{\ell_1} \binom{n}{k_i} - \binom{n-k_t}{k_i} + \sum_{i=\lambda_1+1}^{t} \binom{n-k_t}{k_i-k_t},$$

the upper bound can be achieved if and only if for each $i \in [\ell_1]$, $G_i = \{k_t, n-k_i+2, \ldots, n\}$. □

If for each $i \in [\ell_1]$, $F_i = \{k_t, n-k_i+2, \ldots, n\}$, then for each $i \in [\lambda_1+1, t]$, $F_i = \{1, \ldots, k_t, n-k_i+k_t+1, \ldots, n\}$, and for any L-initial $(n, k_{\ell_1+1}, \ldots, k_{s+1} || k_{s+2}, \ldots, k_{\lambda_1})$-cross intersecting system $(\mathcal{L}(K_{\ell_1+1}, k_{\ell_1+1}), \ldots, \mathcal{L}(K_{\lambda_1}, k_{\lambda_1}))$, we can see that

$$(\mathcal{F}_1, \ldots, \mathcal{F}_{\ell_1}, \mathcal{L}(K_{\ell_1+1}, k_{\ell_1+1}), \ldots, \mathcal{L}(K_{\lambda_1}, k_{\lambda_1}), \mathcal{F}_{\lambda_1+1}, \ldots, \mathcal{F}_t)$$

is a cross intersecting system. Thus, since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal and Claim 4.11, we have $F_i = \{k_t, n-k_i+2, \ldots, n\}$ for each $i \in [\ell_1]$, we are done. □

Denote $t_r = |\{i \in [s+2, t] : k_i = k_{s+2}\}|$. We have the following proposition.

Proposition 4.12. *Under Assumptions 4.4 and 4.7, additionally, suppose that $k_{\ell_{r-1}+1} = \cdots = k_{s+1}$, $n = k_{s+1} + k_{s+2}$ and $s_r \le t_r$. Let $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ be an L-initial extremal $(n, k_1, \ldots, k_{s+1} || k_{s+2}, \ldots, k_t)$-cross intersecting system with ID F_i for each $i \in [t]$. Then for each $i \in [\ell_{r-1}+1, \lambda_{\ell_{r-1}}]$, $F_i = \{1, n-k_i+2, \ldots, n\}$.*

Proof. Considering an extremal L-initial $(n, k_{\ell_{r-1}+1}, \ldots, k_{s+1} || k_{s+2}, \ldots, k_{\lambda_{\ell_{r-1}}})$-cross intersecting system $(\mathcal{G}_{\ell_{r-1}+1}, \ldots, \mathcal{G}_{\lambda_{\ell_{r-1}}})$ with IDs $G_{\ell_{r-1}+1}, \ldots, G_{\lambda_{\ell_{r-1}}}$ that satisfies the conditions of Proposition 4.12. We have the following claim.

Claim 4.13. *For each $i \in [\ell_{r-1}+1, \lambda_{\ell_{r-1}}]$, we have $G_i = \{1, n-k_i+2, \ldots, n\}$.*

Proof. Since $n = k_{s+1} + k_{s+2}$, $|\mathcal{G}_{s+1}| + |\mathcal{G}_{s+2}| \le \binom{n}{k_{s+1}} = \binom{n-1}{k_{s+1}-1} + \binom{n-1}{k_{s+2}-1}$. Since $k_{\ell_{r-1}+1} = \cdots = k_{s+1}$ and $s_r \le t_r$, we have

$$\sum_{i=\ell_{r-1}+1}^{s+1} |\mathcal{G}_i| + \sum_{i=s+2}^{s+1+s_r} |\mathcal{G}_i| \le s_r \binom{n}{k_{s+1}} = s_r \binom{n-1}{k_{s+1}-1} + s_r \binom{n-1}{k_{s+2}-1}.$$

By Assumption 4.7, $s_r \ne \lambda_{\ell_{r-1}} - \lambda_{\ell_r}$. So if $s_r = t_r$, then $t_r < \lambda_{\ell_{r-1}} - \lambda_{\ell_r}$. For each $i \in [s+2+s_r, \lambda_{\ell_{r-1}}]$, we have

$$|\mathcal{G}_i \le \binom{n-1}{k_i-1}.$$

So

$$\sum_{i\in[\ell_{r-1}+1,\lambda_{\ell_{r-1}}]} |\mathcal{G}_i| \le \sum_{i\in[\ell_{r-1}+1,\lambda_{\ell_{r-1}}]} \binom{n-1}{k_i-1},$$

the upper bound can be achieved if and only if for each $i \in [\ell_{r-1}+1, \lambda_{\ell_{r-1}}]$, $G_i = \{1, n-k_i+2, \ldots, n\}$. □

If $F_i = \{1, n-k_i+2, \ldots, n\}$ holds for each $i \in [\ell_{r-1}+1, \lambda_{\ell_{r-1}}]$, then for any L-initial $(n, k_1, \ldots, k_{\ell_{r-1}} || \, k_{\lambda_{\ell_{r-1}}+1}, \ldots, k_t)$-cross intersecting system, say, $(\mathcal{L}(K_1, k_1), \ldots, \mathcal{L}(K_{\ell_{r-1}}, k_{\ell_{r-1}}), \mathcal{L}(K_{\lambda_{\ell_{r-1}}+1}, k_{\lambda_{\ell_{r-1}}+1}), \ldots, \mathcal{L}(K_t, k_t))$, we can see that

$$(\mathcal{L}(K_1, k_1), \ldots, \mathcal{L}(K_{\ell_{r-1}}, k_{\ell_{r-1}}), \mathcal{F}_{\ell_{r-1}+1}, \ldots, \mathcal{F}_{\lambda_{\ell_{r-1}}}, \mathcal{L}(K_{\lambda_{\ell_{r-1}}+1}, k_{\lambda_{\ell_{r-1}}+1}), \ldots, \mathcal{L}(K_t, k_t))$$

is a cross intersecting system. Thus, since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal and Claim 4.13, we have $F_i = \{1, n-k_i+2, \ldots, n\}$ for each $i \in [\ell_{r-1}+1, \lambda_{\ell_{r-1}}]$, we are done. □

The proof of Theorem 4.9 is divided into two cases:

Case I. For each $i \in [s+1]$, $\{1, n-k_i+2, \ldots, n\} \precneqq F_i \precneqq \{k_t, n-k_i+2, \ldots, n\}$.

Case II. There exists $i \in [s+1]$ such that $F_i = \{1, n-k_i+2, \ldots, n\}$ or $F_i = \{k_t, n-k_i+2, \ldots, n\}$.

4.1.1 Case I. For each $i \in [s+1]$, $\{1, n-k_i+2, \ldots, n\} \precneqq F_i \precneqq \{k_t, n-k_i+2, \ldots, n\}$.

In this case, we shall prove a slightly more general result. Let us give some definitions before we state the result. Let k be an integer with $1 \le k \le k_t$. For $i \in [s+1]$, denote

$$\widetilde{\mathbb{R}}_i = \left\{ R \in \binom{n}{k_i} : \{1, n-k_i+2, \ldots, n\} \prec R \prec \{k, n-k_i+2, \ldots, n\} \right\}.$$

For $i \in [s+2, t]$, denote

$$\widetilde{\mathbb{R}}_i = \left\{ R \in \binom{n}{k_i} : \{1, \ldots, k, n-k_i+k+1, \ldots, n\} \prec R \prec \{1, n-k_i+2, \ldots, n\} \right\}.$$

For $j \in [0, r]$, we denote

$$\widetilde{m}_j = \sum_{i=1}^{\ell_j} \left(\binom{n}{k_i} - \binom{n-k}{k_i} \right) + \sum_{i=\lambda_{\ell_j}+1}^{t} \binom{n-k}{k_i-k} + \sum_{i=\ell_j+1}^{\lambda_{\ell_j}} \binom{n-1}{k_i-1}.$$

Denote

$$\widetilde{M}(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t) = \max \Bigg\{ \sum_{i=1}^{t} |\mathcal{A}_i| : (\mathcal{A}_1, \ldots, \mathcal{A}_t) \text{ is an L-initial } (n, k_1, \ldots, k_i \,||\, k_{i+1}, \ldots, k_t)\text{-crossintersecting system with IDs in } \widetilde{\mathbb{R}}_i \Bigg\}.$$

It is clear that for each $i \in [t]$, $\widetilde{\mathbb{R}}_i \subseteq \mathbb{R}_i$. In particular, if $k = k_t$, then $\widetilde{M}(n, k_1, \ldots, k_i || \, k_{i+1}, \ldots, k_t) = M(n, k_1, \ldots, k_i || k_{i+1}, \ldots, k_t)$, $\widetilde{\mathbb{R}}_i = \mathbb{R}_i$ for $i \in [t]$ and $\widetilde{m}_j = m_j$ for $j \in [0, r]$. Clearly, if $k = 1$, then $\sum_{i=1}^{t} |\mathcal{F}_i| = \widetilde{m}_0$. Next, we may assume that

$$k \ge 2.$$

Replacing $[k_t]$ with $[k]$ in Construction 1.5, we get

$$\sum_{i=1}^{t} |\mathcal{F}_i| \ge \max\{\widetilde{m}_0, \ldots, \widetilde{m}_r\}.$$

We will show that Case I will not happen. For this purpose, we are going to prove the following result.

Proposition 4.14. *Under Assumptions 4.4 and 4.7, suppose that $(\widetilde{\mathcal{F}}_1,\dots,\widetilde{\mathcal{F}}_t)$ is an L-initial $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system with IDs $\widetilde{F}_i\in\widetilde{\mathbb{R}}_i$ for each $i\in[t]$. If for each $i\in[s+1]$, $\{1,n-k_i+2,\dots,n\}\precneqq\widetilde{F}_i\precneqq\{k,n-k_i+2,\dots,n\}$, then $(\widetilde{\mathcal{F}}_1,\dots,\widetilde{\mathcal{F}}_t)$ is not extremal, moreover,*

$$\sum_{i=1}^{t}|\widetilde{\mathcal{F}}_i|<\max\{\widetilde{m}_0,\ \widetilde{m}_r\}.$$

The following lemma is crucial, as it enables the reduction of the multi-variable function (9) to a single-value function (see Definition 4.20). We will give the proof in Section 6.

Lemma 4.15. *Under Assumptions 4.4 and 4.7, suppose that $(\widetilde{\mathcal{F}}_1,\dots,\widetilde{\mathcal{F}}_t)$ is an extremal L-initial $(n,k_1,\ \dots,k_{s+1}\,||\,k_{s+2},\dots,k_t)$-cross intersecting system with ID $\widetilde{F}_i\in\widetilde{\mathbb{R}}_i$ for each $i\in[t]$. If for each $i\in[s+1]$, $\{1,n-k_i+2,\dots,n\}\precneqq\widetilde{F}_i\precneqq\{k,n-k_i+2,\dots,n\}$, then for each $i\in[s+1]$, F_i is the corresponding k_i-set of F_1; for each $x\in[r]$, and for each $i\in[\lambda_{\ell_x}+1,\lambda_{\ell_{x-1}}]$, F_i is the k_i-partner of $F_{\ell_{x-1}+1}$.*

The following remark will be frequently used in this paper.

Remark 4.16. *Let $d\geq 2$ be an integer, we consider d L-initial families $\mathcal{L}(A_1,a_1)$, …, $\mathcal{L}(A_d,a_d)$. Fix $i\in[d]$ and let $S\subseteq[d]\setminus\{i\}$ be the set of all $j\in[d]\setminus\{i\}$ satisfying $n\geq a_i+a_j$. Suppose $\{1,n-a_i+2,\dots,n\}\preceq A_i$, then if $\mathcal{L}(A_i,a_i)$ and $\mathcal{L}(A_j,a_j)$ are cross intersecting for each $j\in S$, then for any $j\in S$, every member of $\mathcal{L}(A_j,a_j)$ contains 1. So $\mathcal{L}(A_j,a_j)$, $j\in S$, are pairwise cross intersecting families.*

Combining Fact 2.7 and Remark 4.16, we have the following fact.

Fact 4.17. *Under Assumptions 4.4 and 4.7, let $(\mathcal{G}_1,\dots,\mathcal{G}_t)$ be an L-initial $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$ system with ID G_i for each $i\in[t]$. Suppose that for any $i\in[s]$, either $G_{i+1}\prec G_i$ or G_i is the k_i-parity of G_{i+1}, and for every $j\in[r]$ and every $i\in[\lambda_{\ell_j}+1,\lambda_{\ell_{j-1}}]$, let G_i be the k_i-partner of $G_{\ell_{j-1}+1}$. Then $(\mathcal{G}_1,\dots,\mathcal{G}_t)$ is an $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system.*

Definition 4.18. *Let $R_1\in\mathbb{R}_1$. For every $i\in[2,s+1]$, if R_1 has k_i-parity, then let R_i be the k_i-parity of R_1; otherwise, let R_i be the k_i-set such that $R_i<R_1$. For each $i\in[2,s+1]$, we call R_i the corresponding k_i-set of R_1.*

Fact 4.17 gives the following fact.

Fact 4.19. *Under Assumptions 4.4 and 4.7, let $(\mathcal{G}_1,\dots,\mathcal{G}_t)$ be an L-initial $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$ system with ID G_i for each $i\in[t]$. Suppose that for any $i\in[2,s+1]$, G_i is the corresponding k_i-set of G_1, and for every $j\in[r]$ and every $i\in[\lambda_{\ell_j}+1,\lambda_{\ell_{j-1}}]$, let G_i be the k_i-partner of $G_{\ell_{j-1}+1}$. Then $(\mathcal{G}_1,\dots,\mathcal{G}_t)$ is an $(n,k_1,\dots,k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system.*

Definition 4.20. *Under Assumption 4.7. Let $R_1\in\widetilde{\mathbb{R}}_1$, for each $i\in[2,s+1]$, let R_i be the corresponding k_i-set of R_1; for each $j\in[r]$ and each $i\in[\lambda_{\ell_j}+1,\lambda_{\ell_{j-1}}]$, let R_i be the k_i-partner of $R_{\ell_{j-1}+1}$. Denote*

$$\widetilde{f}(R_1)=\sum_{i=1}^{t}|\mathcal{L}(R_i,k_i)|.$$

Definition 4.21. *For a family $\mathcal{G}\subseteq\widetilde{\mathbb{R}}_1$, denote $\widetilde{f}(\mathcal{G})=\max\{\widetilde{f}(G):G\in\mathcal{G}\}$.*

So to prove Proposition 4.14, it is sufficient to prove the following proposition.

Proposition 4.22. *Suppose that $R_1\in\widetilde{\mathbb{R}}_1$. If $\{1,n-k_1+2,\dots,n\}\precneqq R_1\precneqq\{k,n-k_1+2,\dots,n\}$, then $\widetilde{f}(R_1)<\widetilde{f}(\widetilde{\mathbb{R}}_1)=\max\{\widetilde{m}_0,\widetilde{m}_r\}$.*

Before proving Proposition 4.22, we need to make some preparations first.

Definition 4.23. *Let $R_1, R_1' \in \widetilde{\mathbb{R}}_1$ with $R_1 \prec R_1'$. For every $i \in [2, s+1]$, let R_i, R_i' be the corresponding k_i-sets of R_1, R_1' respectively. For every $j \in [r]$ and every $i \in [\lambda_{\ell_j}+1, \lambda_{\ell_{j-1}}]$, let R_i, R_i' be the k_i-partner of $R_{\ell_{j-1}+1}, R'_{\ell_{j-1}+1}$, respectively. Denote*

$$\Gamma(R_1, R_1') = \sum_{i=1}^{s+1} \big(|\mathcal{L}(R_i', k_i)| - |\mathcal{L}(R_i, k_i)|\big),$$

$$\Delta(R_1, R_1') = \sum_{i=s+2}^{t} \big(|\mathcal{L}(R_i, k_i)| - |\mathcal{L}(R_i', k_i)|\big).$$

For every $i \in [r]$, let

$$\phi_i(R_1, R_1') = \sum_{j=\ell_{i-1}+1}^{\ell_i} \big(|\mathcal{L}(R_j', k_j)| - |\mathcal{L}(R_j, k_j)|\big),$$

$$\chi_i(R_1, R_1') = \sum_{j=\lambda_{\ell_i}+1}^{\lambda_{\ell_{i-1}}} \big(|\mathcal{L}(R_j, k_j)| - |\mathcal{L}(R_j', k_j)|\big).$$

It is easy to see that if $R_1 \prec R_1'$ in $\widetilde{\mathbb{R}}_1$, then

$$\widetilde{f}(R_1') - \widetilde{f}(R_1) = \Gamma(R_1, R_1') - \Delta(R_1, R_1'). \tag{10}$$

Remark 4.24. *Under Assumptions 4.4 and 4.7, for any $j \in [r]$, we have $k_{\ell_{j-1}+1} + k_{\lambda_{\ell_{j-1}+1}+1} \le n < k_{\ell_j - 1} + k_{\ell_j}$. So $n, k_{\ell_{j-1}+1}, \ldots, k_{\ell_j}, k_{\lambda_{\ell_{j-1}+1}+1}, \ldots, k_{\lambda_{\ell_{j-1}}}$ satisfy the conditions in Definition 3.7. In the above definition, for any $R_1 \in \widetilde{\mathbb{R}}_1$ and for any $i \in [2, s+1]$, R_i is the corresponding k_i-set of R_1. Then for any $j \in [r]$ and for any $i \in [\ell_{j-1}+1, \ell_j]$, R_i is the corresponding k_i-set of $F_{\ell_{j-1}+1}$. Thus, for any $j \in [r]$ and $R_1, R_1' \in \widetilde{\mathbb{R}}_1$ with $R_1 \prec R_1'$, we have*

$$\phi_j(R_1, R_1') = \gamma(R_{\ell_{j-1}+1}, R'_{\ell_{j-1}+1})$$

and

$$\chi_j(R_1, R_1') = \delta(R_{\ell_{j-1}+1}, R'_{\ell_{j-1}+1}) = \beta(R_{\ell_{j-1}+1}, R'_{\ell_{j-1}+1}).$$

We will prove the following lemma in Section 5.

Lemma 4.25. *Under Assumptions 4.4 and 4.7, let $A_1, B_1, C_1 \in \mathbb{R}_1$ and a be an integer. Suppose $A_1 \backslash A_1^{\mathrm{t}} = A \cup \{a, a+1\}$, $B_1 = A \cup \{a\} \cup [n - \ell(B_1) + 1, n]$ and $C_1 = A \cup \{a+1\} \cup [n - \ell(B_1) + 1, n]$ for some subset A. For every $i \in [2, s+1]$, let A_i, B_i, C_i be the corresponding k_i-sets of A_1, B_1, C_1 respectively. For every $i \in [r]$ and $j \in [\lambda_{\ell_i}+1, \lambda_{\ell_{i-1}}]$, let A_j, B_j, C_j be the k_j-partners of $A_{\ell_{i-1}+1}, B_{\ell_{i-1}+1}, C_{\ell_{i-1}+1}$ respectively. Then*

(i) $\Gamma(A_1, B_1) \le \Gamma(B_1, C_1)$ and $\Delta(A_1, B_1) \ge \Delta(B_1, C_1)$;

(ii) If $\widetilde{f}(B_1) > \widetilde{f}(A_1)$, then $\widetilde{f}(C_1) > \widetilde{f}(B_1)$.

Now we are going to prove Proposition 4.22.

Proof of Proposition 4.22. The authors have proved the conclusion for the case that $k_1 = k_2 = \cdots = k_{s+1}$ in [12]. So we may assume that there exists $i \in [2, s+1]$ such that $k_i < k_1$. Suppose the assertion of Proposition 4.22 is false, then there exists $R_1 \in \widetilde{\mathbb{R}}_1$ with $\{1, n-k_1+2, \ldots, n\} \precneqq R_1 \precneqq \{k, n-k_1+2, \ldots, n\}$ such that

$$\widetilde{f}(R_1) = \max\{\widetilde{f}(R) : R \in \widetilde{\mathbb{R}}_1\} = \widetilde{f}(\widetilde{\mathbb{R}}_1), \tag{11}$$

We may assume that R_1 is the first member of $\widetilde{\mathbb{R}}_1$ (under lex order) that satisfies the above condition. This implies that for any $R_1' \in \widetilde{\mathbb{R}}_1$ with $\{1, n-k_1+2, \ldots, n\} \precneqq R_1' \precneqq R_1$, $\widetilde{f}(R_1') < \widetilde{f}(\widetilde{\mathbb{R}}_1)$ holds. Suppose that

$$R_1 = A \cup [y+1, y+a] \cup [n-\ell(R_1)+1, n].$$

where $\max A \leq y-1$. We first consider the case that $\ell(R_1) > 0$. Let

$$P_1 = A \cup [y+1, y+a+1] \cup [n-\ell(R_1)+2, n],$$

$$Q_1 = A \cup [y+1, y+a-1] \cup \{y+a+1\} \cup [n-\ell(R_1)+1, n].$$

Clearly, $|P_1| = |Q_1| = k_1$. Since $\ell(R_1) > 0$, P_1, R_1, Q_1 satisfy the conditions of Lemma 4.25. Since $\{1, n-k_1+2, \ldots, n\} \precneqq R_1 \precneqq \{k, n-k_1+2, \ldots, n\}$, $P_1, Q_1 \in \widetilde{\mathbb{R}}_1$ and $\{1, n-k_1+2, \ldots, n\} \precneqq P_1$. By the choice of R_1, we have $\widetilde{f}(P_1) < \widetilde{f}(R_1)$. By Lemma 4.25, we have $\widetilde{f}(R_1) < \widetilde{f}(Q_1)$. A contradiction to (11).

Next, we consider the case $\ell(R_1) = 0$. In this case, let P_1' and Q_1' be two k_1-sets such that $P_1' < R_1 < Q_1'$. Let d be the integer such that $k_1 = \cdots = k_d > k_{d+1}$. For each $j \in [s+1]$, let P_j', R_j, Q_j' be the corresponding k_j-sets of P_1', R_1, Q_1', respectively. For each $j \in [\lambda_1+1, t]$, let P_j', R_j, Q_j' be the k_j-partner of P_1, Q_1, respectively. Following from the definition of corresponding k_j-set, and the conditions $\ell(R_1) = 0$ and $P_1' < R_1$ and case analysis, we have the following claim.

Claim 4.26. *For each $j \in [d]$, $P_j' = P_1'$, $R_j = R_1$ and $Q_j' = Q_1$; for each $j \in [d+1, s+1]$, $P_j' = R_j \prec Q_j'$.*

Since $k_1 = \cdots = k_d$ and $n \geq k_1 + k_{\lambda_1+1}$, we may define α', β' and f_1' functions on the tuple $(n, k_1, \ldots, k_d, k_{\lambda_1+1}, \ldots, k_t)$ to coincide with α, β and f_1 functions on $(n, k_1, \ldots, k_t)$ (see definitions (4), (5, (6))).

Let $\mathbb{S}', \mathbb{S}''$ be systems obtained by replacing each R_j ($j \in [d] \cup [\lambda_1+1, t]$) with P_j, Q_j, respectively. By Claim 4.26, $\mathbb{S}', \mathbb{S}''$ are $(n, k_1, \ldots, k_t)$ cross intersecting systems. Then $f_1'(P_1) \leq f_1'(R_1)$ and $f_1'(Q_1) \leq f_1'(R_1)$ since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal. Since $P_1 < R_1 < Q_1$ and $\max R_1 < n$, $\max R_1 < \max Q_1$. By Proposition 3.3, $\alpha'(P_1, R_1) = \alpha'(R_1, Q_1) = d$, and $\beta'(P_1, R_1) \geq \beta'(R_1, Q_1)$. Thereby, $\beta'(P_1, R_1) = \beta'(R_1, Q_1)$ and $f_1'(P_1) = f_1'(R_1)$. Indeed, if $\beta'(P_1, R_1) > \beta'(R_1, Q_1)$ or $f_1'(P_1) < f_1'(R_1)$, then $f_1'(Q_1) > f_1'(R_1)$, a contradiction. By Proposition 3.3 (iii), $f_1'(P_1) = f_1'(R_1)$ implies $\alpha'(P_1, R_1) = \beta'(P_1, R_1) = d$. Moreover, $\beta'(P_1, R_1) = \beta'(R_1, Q_1) = d > 0$ implies $\beta'(P_1, R_1) = \beta'(R_1, Q_1) = d = t - \lambda_1$ and $n = k_1 + k_{\lambda_1+1} = \cdots = k_1 + k_t$. However, by Proposition 4.10, $F_1 = \{k, n-k_1+2, \ldots, n\}$, a contradiction to the condition of Lemma 4.15. We are done.

So for any $T_1 \in \widetilde{\mathbb{R}}_1$, if $\widetilde{f}(T_1) = \widetilde{f}(\widetilde{\mathbb{R}}_1)$, then $T_1 = \{1, n-k_1+2, \ldots, n\}$ or $T_1 = \{k, n-k_1+2, \ldots, n\}$. If the previous case happens, $\widetilde{f}(\widetilde{\mathbb{R}}_1) = \widetilde{m}_0$, if the latter case happens, then $\widetilde{f}(\widetilde{\mathbb{R}}_1) = \widetilde{m}_r$. The proof of Proposition 4.22 is complete. □

This completes the proof of Case I of Theorem 4.9. As a consequence, we have the following proposition.

Proposition 4.27. *Under Assumptions 4.4 and 4.7, suppose that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an L-initial $(n, k_1, \ldots, k_{s+1} || k_{s+2}, \ldots, k_t)$-cross intersecting system with ID $F_i \in \widetilde{\mathbb{R}}_i$ for each $i \in [t]$. If $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, then $F_1 = \{1, n-k_1+2, \ldots, n\}$ or $F_1 = \{k, n-k_1+2, \ldots, n\}$.*

4.1.2 Case II. There exists $i \in [s+1]$ such that $F_i = \{1, n-k_i+2, \ldots, n\}$ or $F_i = \{k_t, n-k_i+2, \ldots, n\}$

In this section our main result is the following.

Proposition 4.28. *Under Assumptions 4.4 and 4.7, let $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ be an L-initial extremal $(n,k_1,\dots,\ k_{s+1}||k_{s+2},\dots,k_t)$-cross intersecting system with ID F_j for each $j\in[t]$. If there exists $i\in[s+1]$ such that $F_i=\{1,n-k_i+2,\dots,n\}$ or $F_i=\{k_t,n-k_i+2,\dots,n\}$, then*

$$\sum_{i=1}^{t}|\mathcal{F}_i|=\max\{m_0,\dots,m_r\}.$$

Moreover, if $\sum_{i=1}^{t}|\mathcal{F}_i|=m_d$ for some $d\in[r]$, then for each $i\in[\ell_d]$, $F_i=\{k_t,n-k_i+2,\dots,n\}$ and for each $i\in[\ell_d+1,s+1]$, $F_i=\{1,n-k_i+2,\dots,n\}$.

Proof. Let $i\in[s+1]$ be an integer such that $F_i=\{1,n-k_i+2,\dots,n\}$ or $F_i=\{k_t,n-k_i+2,\dots,n\}$ and let $x\in[r]$ be the integer such that $i\in[\ell_{x-1}+1,\ell_x]$. We are going to prove by induction on s.

If $s=0$, then by Theorem 1.1, $\sum_{i=1}^{t}|\mathcal{F}_i|=\max\{m_0,m_1\}$, as desired. Since Assumption 4.7, we will meet Theorem 1.1 (i) and (iii) only. Since $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is L-initial, by Theorem 1.1, if $\sum_{i=1}^{t}|\mathcal{F}_i|=m_1$, then $F_1=\{k_t,n-k_i+2,\dots,n\}$; if $\sum_{i=1}^{t}|\mathcal{F}_i|=m_0$, then $F_1=\{1,n-k_1+2,\dots,n\}$, we are done. We may assume that $s\geq 1$.

We first consider the case that $F_i=\{1,n-k_i+2,\dots,n\}$. By Proposition 4.6, for each $j\in[\ell_{x-1}+1,\lambda_{\ell_{x-1}}]$, $F_j=\{1,n-k_j+2,\dots,n\}$, and $(\mathcal{F}_1,\dots,\mathcal{F}_{\ell_{x-1}},\mathcal{F}_{\lambda_{\ell_{x-1}}+1},\mathcal{F}_t)$ is an extremal L-initial $(n,k_1,\dots,k_{\ell_{x-1}}||k_{\lambda_{\ell_{x-1}}+1},\dots,k_t)$-cross intersecting system.

Let $r'=x-1$, and for each $j\in[r']$, let

$$m'_j=\sum_{y=1}^{\ell_j}\left(\binom{n}{k_y}-\binom{n-k_t}{k_y}\right)+\sum_{y=\lambda_{\ell_j}+1}^{t}\binom{n-k_t}{k_y-k_t}+\sum_{y\in[\ell_j+1,\ell_{x-1}]\cup[\lambda_{\ell_{x-1}+1},\lambda_{\ell_j}]}\binom{n-1}{k_y-1}.$$

By induction hypothesis, we have

$$\sum_{j\in[\ell_{x-1}]\cup[\lambda_{\ell_{x-1}+1},t]}|\mathcal{F}_j|=\max\{m'_0,m'_1,\dots,m'_{r'}\}.$$

If $\max\{m'_0,m'_1,\dots,m'_{r'}\}=m'_d$ for some $d\in[0,r']$, then for each $j\in[\ell_d]$, $F_j=\{k_t,n-k_j+2,\dots,n\}$ and for each $j\in[\ell_d+1,s+1]$, $F_j=\{1,n-k_j+2,\dots,n\}$. Thus, $\sum_{j=1}^{t}|\mathcal{F}_j|=m_d$, as required.

Next, we consider the case that for all $j\in[s+1]$, $\{1,n-k_j+2,\dots,n\}\not\supseteq F_j$ and $F_i=\{k_t,n-k_i+2,\dots,n\}$. Additionally, we may assume that i is the largest number in $[s+1]$ such that $F_i=\{k_t,n-k_i+2,\dots,n\}$. By Proposition 4.6, we get $i=\ell_x$, and

$$|\mathcal{F}_j|=\binom{n}{k_j}-\binom{n-k_t}{k_j}\text{ for each }j\in[\ell_x],$$

$$|\mathcal{F}_j|=\binom{n-k_t}{k_j-k_t}\text{ for each }j\in[\lambda_{\ell_x}+1,t].$$

If $x=r$, then $\sum_{j=1}^{t}|\mathcal{F}_j|=m_r$, we are done. So we next assume that $x<r$. Since $F_i=\{k_t,n-k_i+2,\dots,n\}$ holds for all $i\in[\ell_x]$, $F_j=\{1,\dots,k_t,n-k_j+k_t+1,\dots,n\}$ holds for all $j\in[\lambda_{\ell_x}+1,t]$. Thus, for any $(n,k_{\ell_x+1},\dots,k_{s+1}||k_{s+2},\dots,k_{\lambda_{\ell_x}})$-cross intersecting system with $K_i\in\widetilde{\mathbb{R}}_i{}^{\dagger}$ for each $i\in[\ell_x+1,\lambda_{\ell_x}]$, say, $(\mathcal{L}(K_{\ell_x+1},k_{\ell_x+1}),\dots,\mathcal{L}(K_{\lambda_{\ell_x}},k_{\lambda_{\ell_x}}))$, we can see that the system

$$(\mathcal{F}_1,\dots,\mathcal{F}_{\ell_x},\mathcal{L}(K_{\ell_x+1},k_{\ell_x+1}),\dots,\mathcal{L}(K_{\lambda_{\ell_x}},k_{\lambda_{\ell_x}}),\mathcal{F}_{\lambda_{\ell_x+1}},\dots,\mathcal{F}_t))$$

is cross intersecting. So $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ being extremal implies that $(\mathcal{L}(F_{\ell_x+1},k_{\ell_x+1}),\dots,\mathcal{L}(F_{\lambda_{\ell_x}},k_{\lambda_{\ell_x}}))$ is also extremal satisfying that $F_i\in\widetilde{\mathbb{R}}_i$ for each $i\in[\ell_x+1,\lambda_{\ell_x}]$. However, for each $j\in[\ell_x+1,\lambda_{\ell_x}]$,

[†]Here, we need take $k=k_t$ in the definition of $\widetilde{\mathbb{R}}_i$.

neither $F_j = \{1, n-k_j+2, \ldots, n\}$ nor $F_j = \{k_t, n-k_j+2, \ldots, n\}$. By Proposition 4.14, $(\mathcal{L}(F_{\ell_x+1}, k_{\ell_x+1}), \ldots, \mathcal{L}(F_{\lambda_{\ell_x}}, k_{\lambda_{\ell_x}}))$ is not extremal on $\widetilde{\mathbb{R}}_i$, a contradiction.

This completes the proof of Proposition 4.28. □

We have finished the proof of the quantitative part of Therorem 4.9. Next we consider the characterization of extremal systems in Theorem 4.9.

4.2 Characterization of extremal systems in Theorem 4.9

Huang, Peng and Wang [11] proved the following result, which will be used later.

Lemma 4.29 (Corollary 2.3 in [11]). *Let $\mathcal{A} \subseteq \binom{[n]}{a}$ and $\mathcal{B} \subseteq \binom{[n]}{b}$ be cross intersecting families with $n > a+b$. If $|\mathcal{A}| = \binom{n-1}{a-1}$ and $\cap\{A\colon A \in \mathcal{A}\} = \emptyset$, then $|\mathcal{B}| < \binom{n-1}{b-1}$.*

Let $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ be an extremal $(n, k_1, \ldots, k_{s+1} || k_{s+2}, \ldots, k_t)$-cross intersecting system satisfying Assumptions 4.4 and 4.7. So

$$\sum_{i=1}^{t} |\mathcal{F}_i| = \max\{m_0, m_1, \ldots, m_r\}.$$

Let $j \in [0, r]$ be the integer such that $\sum_{i=1}^{t} |\mathcal{F}_i| = m_j$. We are going to prove that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is isomorphic to the system $(\mathcal{A}_{j,1}, \mathcal{A}_{j,2}, \ldots, \mathcal{A}_{j,t})$ in Construction 1.5.

By Propositions 4.27 and 4.28, for each $i \in [\ell_j]$, we have $|\mathcal{F}_i| = \binom{n}{k_i} - \binom{n-k_t}{k_i}$, and for each $i \in [\ell_j+1, s+1]$, we have $|\mathcal{F}_i| = \binom{n-1}{k_i-1}$. Note that for each $i \in [\ell_j]$, $n \geq k_i + k_t$, $\mathcal{F}_i$ and $\mathcal{F}_t$ are cross intersecting families and $\mathcal{F}_t \neq \emptyset$. Let $F \in \mathcal{F}_t$. Then for any $G \in \mathcal{F}_i$, $G \cap F \neq \emptyset$. Since $|F| = k_t$, then $|\mathcal{F}_i| \leq \binom{n}{k_i} - \binom{n-k_t}{k_i}$, with equality holds if and only if $\mathcal{F}_t = \{F\}$. So for each $i \in [\ell_j]$, $\mathcal{F}_i = \left\{A \in \binom{[n]}{k_i} : A \cap F \neq \emptyset\right\}$, which is isomorphic to $\mathcal{A}_{j,i}$, as required. This implies that for each $i \in [\lambda_{\ell_j}+1, t]$, $\mathcal{F}_i = \left\{A \in \binom{[n]}{k_i} : F \subseteq A\right\}$, which is isomorphic to $\mathcal{A}_{j,i}$, as required. Note that $\sum_{i=1}^{t} |\mathcal{F}_i| = m_j$, $|\mathcal{F}_i| = \binom{n}{k_i} - \binom{n-k_t}{k_i}$ for each $i \in [\ell_j]$, $|\mathcal{F}_i| = \binom{n-k_t}{k_i-k_t}$ for each $i \in [\lambda_{\ell_j}+1, t]$, and $|\mathcal{F}_i| = \binom{n-1}{k_i-1}$ for each $i \in [\ell_j+1, s+1]$. By the Kruskal–Katona theorem, we obtain that for each $i \in [s+2, \lambda_{\ell_j}]$, $|\mathcal{F}_i| = \binom{n-1}{k_i-1}$. If $j = r$, then we are done. Assume that $j \leq r-1$. Then $s+2 \leq \lambda_{\ell_j}$. We first consider the case: $s+2 < \lambda_{\ell_j}$. In this case, for any i, h in $[s+2, \lambda_{\ell_j}]$ with $i \neq h$, we have $n > k_i + k_h$. Then by Lemma 4.29, we obtain that there exists $a \in F$ such that for each $i \in [s+2, \lambda_{\ell_j}]$, $\mathcal{F}_i = \left\{A \in \binom{[n]}{k_i} : a \in A\right\}$. Then for each $i \in [\ell_j+1, s+1]$, we have $\mathcal{F}_i = \left\{A \in \binom{[n]}{k_i} : a \in A\right\}$, as required. Next, we consider the case $s+2 = \lambda_{\ell_j}$. We are going to confirm that this case will not happen. In fact, $s+2 = \lambda_{\ell_j}$ implies $j = r-1$. We only need to consider the cross intersecting families $\mathcal{F}_{\ell_{r-1}+1}, \ldots, \mathcal{F}_{s+1}, \mathcal{F}_{s+2}$. If $n = k_{s+1} + k_{s+2}$, then $k_{\ell_{r-1}+1} = \cdots = k_{s+1}$, so $s_r \geq 2$ (recall the Assumption4.7). Since $k_t \geq 2$, we get $\sum_{i\in[\ell_{r-1}+1, s+2]} |\mathcal{F}_i| = \sum_{i\in[\ell_{r-1}+1, s+2]} \binom{n-1}{k_i-1} < \sum_{i\in[\ell_{r-1}+1, s+1]} \left(\binom{n}{k_i} - \binom{n-k_t}{k_i}\right) + \binom{n-k_t}{k_{s+2}-k_t}$. This implies $\sum_{i\in[t]} |\mathcal{F}_i| < m_j$, a contradiction. We are done.

4.3 Proofs of Theorems 1.4 and 1.7

In this section, we will use the following facts.

Fact 4.30. *Let $n \geq a+b$ and $a > b$. Then $\binom{n}{a} \geq \binom{n}{b}$, moreover, equality holds if and only if $n = a+b$.*

Proof. If $b < a \leq \frac{n}{2}$, then by monotonicity, we are done. Thus we may assume that $b \leq \frac{n}{2} < a$. Then $n \geq a+b$ implies that $\frac{n}{2} > n-a \geq b$. It follows that $\binom{n}{a} = \binom{n}{n-a} \geq \binom{n}{b}$. Moreover, the equality holds if and only if $n = a+b$, as desired. □

Fact 4.31. *Let n, a, b, c be integers with $n \geq a+b+1$ and $a \geq b \geq c$. Then*

$$\sum_{j=2}^{c} \binom{n-j}{a-1} \geq \sum_{j=2}^{c} \binom{n-j}{b-1}. \tag{12}$$

Proof. Note that $\sum_{j=2}^{c} \binom{n-j}{a-1} = \binom{n-1}{a} - \binom{n-c}{a}$ and $\sum_{j=2}^{c} \binom{n-j}{b-1} = \binom{n-1}{b} - \binom{n-c}{b}$. By Fact 4.30, $\binom{n-1}{a} \geq \binom{n-1}{b}$. If $\binom{n-c}{a} \leq \binom{n-c}{b}$, then we are done. Suppose that $\binom{n-c}{a} > \binom{n-c}{b}$. Since $a \geq b$, $b \leq \frac{n-c}{2}$. We claim that $\binom{n-c}{a-1} > \binom{n-c}{b-1}$. Indeed, if $b-1 \leq a-1 \leq \frac{n-c}{2}$, then we are done. If $b-1 \leq \frac{n-c}{2} \leq a-1$, then since $\binom{n-c}{a} > \binom{n-c}{b}$, we have $\binom{n-c}{a-1} \geq \binom{n-c}{a} > \binom{n-c}{b} > \binom{n-c}{b-1}$. Similarly, $\binom{n-c}{a-2} > \binom{n-c}{b-2}$. So $\binom{n-c+1}{a-1} - \binom{n-c+1}{b-1} = \binom{n-c}{a-1} + \binom{n-c}{a-2} - \binom{n-c}{b-1} - \binom{n-c}{b-2} > 0$. Similarly, $\binom{n-c+j}{a-1} > \binom{n-c+j}{b-1}$ holds for any $j \geq 0$, yielding $\sum_{j=2}^{c} \binom{n-j}{a-1} > \sum_{j=2}^{c} \binom{n-j}{b-1}$, as desired. □

Denote

$$I = \{i \in [r] : s_i = \lambda_{\ell_{i-1}} - \lambda_{\ell_i}, k_{\ell_{i-1}+1} = \cdots = k_{\ell_i}, k_{\lambda_{\ell_i}+1} = \cdots = k_{\lambda_{\ell_{i-1}}}, n = k_{\ell_i} + k_{\lambda_{\ell_i}+1}\}.$$

The proof will be divided into the following two cases.

Case 1. $I = \emptyset$.

Since $I = \emptyset$, Assumptions 4.4 and 4.7 are satisfied. By Theorem 4.9, it is sufficient to prove the following proposition.

Proposition 4.32. *Let $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ be an L-initial extremal $(n, k_1, \ldots, k_t)$-cross intersecting system satisfying the conditions of Theorem 1.4. Suppose that $I = \emptyset$, then $|\mathcal{F}_i| \geq \binom{n-1}{k_i-1}$ holds for every $i \in [w+1]$.*

Proof. To prove this proposition, we only need to confirm that $\{1, n-k_{w+1}+2, \ldots, n\} \prec F_{w+1}$. Indeed, if $\{1, n-k_{w+1}+2, \ldots, n\} \prec F_{w+1}$, then since $n \geq k_{w+1} + k_{w+2}$, $k_1 \geq k_2 \geq \cdots \geq k_t$, and $\mathcal{F}_{w+1}$ and $\mathcal{F}_j$ ($j \in [w+2, t]$) are non-freely cross intersecting, we have $F_j \prec \{1, n-k_j+2, \ldots, n\}$ for all $j \in [w+2, t]$. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, $\{1, n-k_i+2, \ldots, n\} \prec F_i$ for all $i \in [w+1]$, we are done.

Suppose on the contrary that $F_{w+1} \precneqq \{1, n-k_{w+1}+2, \ldots, n\}$. We will show this leads to a contradiction by induction on w. If $w = 0$, then by Theorem 1.1, we are done. (The only nontrivial case is Theorem 1.1 (ii), however, since $I = \emptyset$, we will not meet this case.) Next, we consider $w \geq 1$.

We first consider the case: $s_r > 1$. In this case, $\ell_{r-1}+1 \neq w+1$, and $(\mathcal{F}_1, \ldots, \mathcal{F}_w, \mathcal{F}_{w+2}, \ldots, \mathcal{F}_t)$ is extremal. By induction hypothesis on $(\mathcal{F}_1, \ldots, \mathcal{F}_w, \mathcal{F}_{w+2}, \ldots, \mathcal{F}_t)$, then for each $i \in [w]$, $\{1, n-k_i+2, \ldots, n\} \prec F_i$. Particularly, $\{1, n-k_{\ell_{r-1}+1}+2, \ldots, n\} \prec F_{\ell_{r-1}+1}$. For all $i \in [w+2, t]$, $n \geq k_{\ell_{r-1}+1} + k_i$. In view of Remark 4.16, $F_i \prec \{1, n-k_i+2, \ldots, n\}$ for all $i \in [w+2, t]$. Thus, if we replace F_{w+1} by $\{1, n-k_{w+1}+2, \ldots, n\}$, the new system is still cross intersecting, this makes a contradiction to the fact that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal.

Next we consider the case that $s_r = 1$. Since we've proved the case $w = 0$, if $r = 1$, we are done. We may assume that $r \geq 2$.

First, we consider the case: for each $i \in [\ell_{r-1}]$, $F_i = \{k_t, n-k_i+2, \ldots, n\}$. Therefore, in this case, cross intersecting system $(\mathcal{F}_{w+1}, \ldots, \mathcal{F}_{\lambda_{\ell_{r-1}}})$ is extremal. By Theorem 1.1 again, $\{1, n-k_{w+1}+2, \ldots, n\} \prec F_{w+1}$, we are done. Thus, next, we may assume that there is $k \in [\ell_{r-1}]$ such that $F_k \precneqq \{k_t, n-k_k+2, \ldots, n\}$.

Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, there is an $i \in [w+2, t]$ such that $\{1, n-k_i+2, \ldots, n\} \prec F_i$. (Otherwise, $\{1, n-k_i+2, \ldots, n\} \prec F_i$ for every $i \in [w+1]$ since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal.) Let $x \in [r]$ be the integer such that $i \in [\lambda_{\ell_x}+1, \lambda_{\ell_{x-1}}]$. Then $F_j \prec \{1, n-k_j+2, \ldots, n\}$ for all $j \in [\ell_{x-1}+1, t] \setminus \{i\}$. Therefore, $\{1, n-k_j+2, \ldots, n\} \prec F_j$ for $j \in [\ell_{x-1}] \cup \{i\}$ Note that

$$(\mathcal{F}_1, \ldots, \mathcal{F}_{\ell_{x-1}}, \mathcal{F}_i, \mathcal{F}_{\ell_{x-1}+1}, \ldots, \mathcal{F}_{i-1}, \mathcal{F}_{i+1}, \ldots, \mathcal{F}_t)$$

is an $(k_1, k_2, \ldots, k_{\ell_{x-1}}, k_i || k_{\ell_{x-1}+1}, \ldots, k_{i-1}, k_{i+1}, \ldots, k_t)$ cross system. We will apply Theorem 4.9 to this system. For this purpose, we define the following numbers.

For $j \in [0, x-1]$, we denote

$$m_j(x) = \sum_{d=1}^{\ell_j} \left(\binom{n}{k_d} - \binom{n-k_t}{k_d} \right) + \sum_{d=\lambda_{\ell_j}+1}^{t} \binom{n-k_t}{k_d-k_t} + \sum_{d=\ell_j+1}^{\lambda_{\ell_j}} \binom{n-1}{k_d-1}.$$

If $i \neq t$, then we denote

$$m_x(x) = \sum_{d=1}^{\ell_{x-1}} \left(\binom{n}{k_d} - \binom{n-k_t}{k_d} \right) + \binom{n}{k_i} - \binom{n-k_t}{k_i} + \sum_{d \in [t] \setminus ([\ell_{x-1}] \cup \{i\})} \binom{n-k_t}{k_d-k_t},$$

If $i = t$, we have $x = 1$, then we denote

$$m_1(1) = \binom{n}{k_t} - \binom{n-k_{t-1}}{k_t} + \sum_{d=1}^{t-1} \binom{n-k_{t-1}}{k_d-k_{t-1}}.$$

Applying Theorem 4.9 to the $(k_1, k_2, \ldots, k_{\ell_{x-1}}, k_i || k_{\ell_{x-1}+1}, \ldots, k_{i-1}, k_{i+1}, \ldots, k_t)$ cross system $(\mathcal{F}_1, \ldots, \mathcal{F}_{\ell_{x-1}}, \mathcal{F}_i, \mathcal{F}_{\ell_{x-1}+1}, \ldots, \mathcal{F}_{i-1}, \mathcal{F}_{i+1}, \ldots, \mathcal{F}_t)$, we have

$$\sum_{j=1}^{t} |\mathcal{F}_j| = \max\{m_0(x), m_1(x), \ldots, m_{x-1}(x), m_x(x)\}.$$

By Theorem 4.9 (the 'equality holds' part), if $j \in [0, x-1]$, then $F_{w+1} = \{1, n-k_{w+1}+2, \ldots, n\}$, we are done. Thus

$$\sum_{j=1}^{t} |\mathcal{F}_j| = m_x(x) > \max\{m_0(x), m_1(x), \ldots, m_{x-1}(x)\}, \tag{13}$$

and for all $j \in [\ell_{x-1}] \cup \{i\}$, we have $F_j = \{k_t, n-k_j+2, \ldots, n\}$. By the assumption that there is $k \in [\ell_{r-1}]$ such that $F_k \not\supseteq \{k_t, n-k_k+2, \ldots, n\}$, we have $x \leq r-1$. We have the following claim.

Claim 4.33. $m_x > m_x(x)$.

Proof. We first suppose $i = t$. By Proposition 3.4, we have

$$\binom{n}{k_t} - \binom{n-k_{t-1}}{k_t} + \sum_{j=1}^{t-1} \binom{n-k_{t-1}}{k_j-k_{t-1}} \leq \max\left\{ \binom{n}{k_1} - \binom{n-k_t}{k_1} + \sum_{j=2}^{t} \binom{n-k_t}{k_j-k_t}, m_0(1) \right\}.$$

By (13), we have $m_1(1) \leq \binom{n}{k_1} - \binom{n-k_t}{k_1} + \sum_{d=2}^{t} \binom{n-k_t}{k_d-k_t}$. Since $k_t \geq 2$,

$$m_1 - m_1(1) \geq \sum_{d=2}^{\lambda_{l_1}} \binom{n-1}{k_d-1} - \binom{n-k_t}{k_d-k_t} > 0,$$

as required.

Next we assume that $i \leq t-1$. Since $x \leq r-1$,

$$m_x - m_x(x) > \binom{n}{k_{\ell_x}} - \binom{n-k_t}{k_{\ell_x}} + \binom{n-k_t}{k_i-k_t} - \left(\binom{n}{k_i} - \binom{n-k_t}{k_i} + \binom{n-k_t}{k_{\ell_x}-k_t} \right).$$

We claim that

$$\binom{n}{k_{\ell_x}} - \binom{n-k_t}{k_{\ell_x}} - \binom{n-k_t}{k_{\ell_x}-k_t} \geq \binom{n}{k_i} - \binom{n-k_t}{k_i} - \binom{n-k_t}{k_i-k_t}. \tag{14}$$

Proof of (14). If $n = k_i + k_{\ell_x}$, then

$$\binom{n}{k_i}-\binom{n-k_t}{k_i}-\binom{n-k_t}{k_i-k_t}=\binom{n}{k_i}-\binom{n-k_t}{k_i}-\binom{n-k_t}{k_{\ell_x}}=\binom{n}{k_{\ell_x}}-\binom{n-k_t}{k_{\ell_x}}-\binom{n-k_t}{k_{\ell_x}-k_t},$$

we are fine. Next we may assume that $n \geq k_{\ell_x}+k_i+1$. Then for any $2 \leq p \leq k_t$, Fact 4.30 gives

$$\binom{n-p}{k_i-p+1} < \binom{n-p}{k_{\ell_x}-p+1}. \tag{15}$$

Let explain why equality does not hold in (15). Suppose on the contrary that there exists $p \in [2, k_t]$ such that $\binom{n-p}{k_i-p+1} = \binom{n-p}{k_{\ell_x}-p+1}$. Since $k_i < k_{\ell_x}$, $k_i - p + 1 < k_{\ell_x} - p + 1$. By Fact 4.30, $\binom{n-p}{k_i-p+1} = \binom{n-p}{k_{\ell_x}-p+1}$ implies $n - p = k_i - p + 1 + k_{\ell_x} - p + 1$. Then $n = k_i + k_{\ell_x} - p + 2$. Since $n \geq k_{\ell_x} + k_i + 1$, $p \leq 1$. This is a contradiction to $p \geq 2$. Thus,

$$\begin{aligned}\binom{n}{k_i}-\binom{n-k_t}{k_i}-\binom{n-k_t}{k_i-k_t} &= \sum_{2\leq p\leq k_t}\binom{n-p}{k_i-1}+\sum_{2\leq p\leq k_t}\binom{n-p}{k_i-p+1}\\ &\overset{\text{by (12),(15)}}{<} \sum_{2\leq p\leq k_t}\binom{n-p}{k_{\ell_x}-1}+\sum_{2\leq p\leq k_t}\binom{n-p}{k_{\ell_x}-p+1}\\ &= \binom{n}{k_{\ell_x}}-\binom{n-k_t}{k_{\ell_x}}-\binom{n-k_t}{k_{\ell_x}-k_t},\end{aligned}$$

as required. □

This proves Claim 4.33. □

Clearly, Claim 4.33 makes a contradiction to the fact that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal. This proves Proposition 4.32. □

Case 2. $I \neq \emptyset$.

If $I = [r]$, then we are done by Proposition 4.2. So we may assume that $[r] \setminus I \neq \emptyset$. Let

$$I = \{i_1, i_2, \ldots, i_c\}, [r] \setminus I = \{j_1, \ldots, j_d\}.$$

Let

$$\begin{aligned}X &= \cup_{j\in[c]}\{\ell_{i_j-1}+1, \ldots, \ell_{i_j}, \lambda_{\ell_{i_j}}+1, \ldots, \lambda_{\ell_{i_j-1}}\} = \{e_1, \ldots, e_a\},\\ Y &= [t] \setminus X = \{f_1, \ldots, f_b\}.\end{aligned}$$

By Proposition 4.2 and the definition of I, we have the following claim.

Claim 4.34.

$$\sum_{i=1}^{a}|\mathcal{F}_{e_i}| \leq \sum_{i=1}^{a}\binom{n-1}{k_{e_i}-1},$$

equality holds if and only if $(\mathcal{F}_{e_1}, \ldots, \mathcal{F}_{e_a})$ is isomorphic to the families as defined in Construction 1.6 (replacing $1, \ldots t$ in Construction 1.6 with $e_1, \ldots, e_a$ is needed).

Since

$$\begin{aligned}\sum_{i=1}^{b}|\mathcal{F}_{f_i}| &\leq M(n, k_{f_1}, \ldots, k_{f_b}),\\ \sum_{i=1}^{a}|\mathcal{F}_{e_i}| &\leq M(n, k_{e_1}, \ldots, k_{e_a}),\end{aligned} \tag{16}$$

we have

$$\sum_{i=1}^{t}|\mathcal{F}_i| \le M(n,k_{e_1},\dots,k_{e_a}) + M(n,k_{f_1},\dots,k_{f_b}). \tag{17}$$

We are going to prove that the right side of (17) is reachable.

For each $i \in [d]$, we denote m'_i analogously as we define m_j in (2).

By **Case 1**, there exists $p \in [0,d]$ such that

$$M(n,k_{f_1},\dots,k_{f_b}) = m'_p,$$

To make the equality holds in (16), by **Case 1**, there exist a k_t-set F and a number $x \in F$ such that for each $g \in [p]$ (here $p > 0$ is needed), and for each $h \in [\ell_{j_g-1}+1, \ell_{j_g}]$,

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : A \cap F \neq \emptyset \right\}, \tag{18}$$

for each $g \in [p]$, and for each $h \in [\lambda_{\ell_{j_g}}+1, \lambda_{\ell_{j_g-1}}]$,

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : F \subseteq A \right\}, \tag{19}$$

for each $g \in [p+1,d]$ (here $p < d$ is needed), and for each $h \in [\ell_{j_g-1}+1, \ell_{j_g}] \cup [\lambda_{\ell_{j_g}}+1, \lambda_{\ell_{j_g-1}}]$,

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : x \in A \right\}. \tag{20}$$

The following claim is derived from Proposition 4.6 directly.

Claim 4.35. *(i) If $p > 0$, then for each $g \in [c]$ with $i_g < j_p$, for each $h \in [\ell_{i_g-1}+1, \ell_{i_g}]$, we have $\mathcal{F}_h = \{A \in \binom{[n]}{k_h} : A \cap F \neq \emptyset\}$ and $\mathcal{F}_{\lambda_h+1} = \{A \in \binom{[n]}{k_{\lambda_h+1}} : F \subseteq A\}$.*

(ii) If $p < d$, then for each $g \in [c]$ with $i_g > j_{p+1}$, for each $h \in [\ell_{i_g-1}+1, \ell_{i_g}]$, we have $\mathcal{F}_h = \{A \in \binom{[n]}{k_h} : x \in A\}$ and $\mathcal{F}_{\lambda_h+1} = \{A \in \binom{[n]}{k_{\lambda_h+1}} : x \in A\}$.

Consider the following three cases.

(a) $p = 0$, and there is $g \in [c]$ such that $i_g < j_1$.

(b) $p = d$, and there is $g \in [c]$ such that $i_g > j_p$.

(c) $0 < p < d$, and there is $g \in [c]$ such that $j_p < i_g < j_{p+1}$.

If none of the above three cases happen, then by Claim 4.35, we get that $\sum_{i=1}^{t}|\mathcal{F}_i| = m_{j_p}$ (here $j_0 = 0$) and for each $i \in [t]$, $\mathcal{F}_i$ is isomorphic to $\mathcal{A}_{j_p,i}$. This corresponds to Theorem 1.7 (i).

Next, we assume that one of the above cases happens.

(a) Suppose that $p = 0$, and there is $g \in [c]$ such that $i_g < j_1$.

Let T be the set of integers g satisfying **Case (a)**. Then $T = \{1,\dots,j_1-1\}$. Combining (20) and Claim 4.35 (ii), we obtain that

$$\mathcal{F}_h = \{A \in \binom{[n]}{k_h} : x \in A\} \text{ for each } h \in [\ell_{j_1-1}+1, \lambda_{\ell_{j_1-1}}]. \tag{21}$$

Recall that $(\mathcal{F}_1,\dots,\mathcal{F}_t)$ is extremal. And (21) guarantees that $(\mathcal{F}_1,\dots,\mathcal{F}_{\ell_{j_1-1}},\mathcal{F}_{\lambda_{\ell_{j_1-1}}+1},\dots,\mathcal{F}_t)$ is extremal. Recalling the definition of I, and applying Proposition 4.2, we obtain that for all $h \in [\ell_{j_1-1}] \cup [\lambda_{\ell_{j_1-1}}+1, t]$, the families $\mathcal{F}_h$ are isomorphic to the families in Construction 1.6 (replacing 1 by x and $[k_t]$ by F), and $\sum_{i=1}^{t}|\mathcal{F}_i| = m_0$. We meet Theorem 1.7 (ii).

(b) Suppose that $p = d$, and there is $g \in [c]$ such that $i_g > j_d$.

Let T be the set of integers g satisfying $i_g > j_d$. Then $T = \{j_d + 1, \ldots, r\}$. Combining (18), (19) and Claim 4.35 (i), we obtain that

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : A \cap F \neq \emptyset \right\} \text{ for each } h \in [\ell_{j_p}], \tag{22}$$

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : F \subseteq A \right\} \text{ for each } h \in [\lambda_{\ell_{j_p}} + 1, t]. \tag{23}$$

Recall that $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal. And (22), (23) guarantee that $(\mathcal{F}_{\ell_{j_p}+1}, \ldots, \mathcal{F}_{\lambda_{\ell_{j_p}}})$ is extremal. Recall the definition of I, and applying Proposition 4.2, we obtain that for all $h \in [\ell_{j_p}+1, \lambda_{\ell_{j_p}}]$, families $\mathcal{F}_h$ are isomorphic to the families in Construction 1.6, and $\sum_{i=1}^{t} |\mathcal{F}_i| = m_{j_p}$. We meet Theorem 1.7 (iii).

(c) Suppose that $0 < p < d$, and there is $g \in [c]$ such that $j_p < i_g < j_{p+1}$.

Let T be the set of integers g satisfying $j_p < i_g < j_{p+1}$. Then $T = \{j_p + 1, \ldots, j_{p+1} - 1\}$. Combining (18), (19), (20) and Claim 4.35, we obtain that

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : A \cap F \neq \emptyset \right\} \text{ for each } h \in [\ell_{j_p}],$$

$$\mathcal{F}_h = \left\{ A \in \binom{[n]}{k_h} : F \subseteq A \right\} \text{ for each } h \in [\lambda_{\ell_{j_p}} + 1, t],$$

$$\mathcal{F}_h = \{ A \in \binom{[n]}{k_h} : x \in A \} \text{ for each } h \in [\ell_{j_{p+1}-1} + 1, \lambda_{\ell_{j_{p+1}-1}}].$$

By a similar argument as above, We obtain that for all $h \in [\ell_{j_p}+1, \ell_{j_{p+1}-1}] \cup [\lambda_{\ell_{j_p+1}}+1, \lambda_{\ell_{j_{p+1}-1}}]$, the families $\mathcal{F}_h$ are isomorphic to the families in Construction 1.6, and $\sum_{i=1}^{t} |\mathcal{F}_i = m_{j_p}$. We meet Theorem 1.7 (iv). This concludes the proofs of Theorems 1.4 and 1.7.

5 Proof of Lemma 4.25

For every $i \in [2, s+1]$, let A_i, B_i, C_i be the corresponding k_i-sets of A_1, B_1, C_1, respectively. For every $j \in [r]$ and every $i \in [\lambda_{\ell_j}+1, \lambda_{\ell_{j-1}}]$, let A_i, B_i, C_i be the k_i-partner of $A_{\ell_{j-1}+1}, B_{\ell_{j-1}+1}, C_{\ell_{j-1}+1}$, respectively.

We claim that Lemma 4.25 (ii) follows from Lemma 4.25 (i). Let's explain this. By (10), we have $\widetilde{f}(B_1) - \widetilde{f}(A_1) = \Gamma(A_1, B_1) - \Delta(A_1, B_1)$ and $\widetilde{f}(C_1) - \widetilde{f}(B_1) = \Gamma(B_1, C_1) - \Delta(B_1, C_1)$. Since $\widetilde{f}(B_1) > \widetilde{f}(A_1)$, $\Gamma(A_1, B_1) > \Delta(A_1, B_1)$. Combining this with Lemma 4.25 (i), we have $\Gamma(B_1, C_1) > \Delta(B_1, C_1)$, therefore, $\widetilde{f}(C_1) > \widetilde{f}(B_1)$. We get Lemma 4.25 (ii).

We are going to prove Lemma 4.25 (i) by induction on r. If $r = 1$, then $n, k_1, \ldots, k_{s+1}$, $k_{s+2}, \ldots, k_t$ (here, the number $s+1$ corresponds to the number s in Definition 3.7) satisfy the condition of Definition 3.7. So A_1, B_1, C_1 satisfy the condition of Lemma 3.9. From Definition 4.23 and Remark 4.24, we have $\Gamma(A_1, B_1) = \gamma(A_1, B_1)$ and $\Delta(A_1, B_1) = \delta(A_1, B_1) = \beta(A_1, B_1)$. Analogous expressions hold for (B_1, C_1). By Lemma 3.9, we have $\delta(A_1, B_1) = \delta(B_1, C_1)$ and $\gamma(A_1, B_1) \leq \gamma(B_1, C_1)$. We get Lemma 4.25 (i). Next, we assume that $r \geq 2$ and Lemma 4.25 holds for $r - 1$.

Claim 5.1. *$\phi_r(A_1, B_1) \leq \phi_r(B_1, C_1)$ and $\chi_r(A_1, B_1) \geq \chi_r(B_1, C_1)$.*

Proof. By Remark 4.24, we have

$$\phi_r(A_1, B_1) = \gamma(A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}),$$
$$\chi_r(A_1, B_1) = \delta(A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}) = \beta(A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}).$$

Analogous expressions hold for (B_1, C_1). Since $A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}, C_{\ell_{r-1}+1}$ are the corresponding $k_{\ell_{r-1}+1}$-sets of A_1, B_1, C_1, we will meet one of the following three cases:

(a) $A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}, C_{\ell_{r-1}+1}$ satisfy the condition of Lemma 3.9. (Here, $A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}, C_{\ell_{r-1}+1}$ correspond to A_1, B_1, C_1 in the statement of Lemma 3.9.)

(b) $A_{\ell_{r-1}+1} = B_{\ell_{r-1}+1} = C_{\ell_{r-1}+1}$ in $\widetilde{\mathbb{R}}_{\ell_{r-1}+1}$;

(c) $A_{\ell_{r-1}+1} < B_{\ell_{r-1}+1} < C_{\ell_{r-1}+1}$ in $\widetilde{\mathbb{R}}_{\ell_{r-1}+1}$.

Clearly, case (b) immediately yields the desired result. Since the case $r = 1$ has already been established, the same conclusion holds if (a) occurs. We therefore assume that (c) is the only remaining possibility. In this case, for any $i \in [\ell_{r-1}+1, s+1]$, if $k_i < k_{\ell_{r-1}+1}$, then $A_i = B_i \prec C_i$; if $k_i = k_{\ell_{r-1}+1}$, then $A_i < B_i < C_i$ in $\widetilde{\mathbb{R}}_{\ell_{r-1}+1}$. Thus, $\phi_r(A_1, B_1) \leq \phi_r(B_1, C_1)$, as required. Since $A_{\ell_{r-1}+1} < B_{\ell_{r-1}+1} < C_{\ell_{r-1}+1}$ in $\widetilde{\mathbb{R}}_{\ell_{r-1}+1}$, $\max B_{\ell_{r-1}+1} \leq \max C_{\ell_{r-1}+1}$. Since $\chi_r(A_1, B_1) = \beta(A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1})$, by Proposition 3.3, we have

$$\chi_r(A_1, B_1) = \beta(A_{\ell_{r-1}+1}, B_{\ell_{r-1}+1}) \geq \beta(B_{\ell_{r-1}+1}, C_{\ell_{r-1}+1}) = \chi_r(B_1, C_1),$$

we are done. □

Recall Definition 4.23. Combining the induction hypothesis and Claim 5.1, we have

$$\begin{aligned}\Delta(A_1, B_1) &= \sum_{i=1}^{r-1} \chi_i(A_1, B_1) + \chi_r(A_1, B_1) \\ &\leq \sum_{i=1}^{r-1} \phi_i(B_1, C_1) + \phi_r(B_1, C_1) \\ &= \Gamma(B_1, C_1),\end{aligned}$$

and

$$\begin{aligned}\Gamma(A_1, B_1) &= \sum_{i=1}^{r-1} \phi_i(A_1, B_1) + \phi_r(A_1, B_1) \\ &\geq \sum_{i=1}^{r-1} \chi_i(B_1, C_1) + \chi_r(B_1, C_1) \\ &= \Delta(B_1, C_1),\end{aligned}$$

as required. This concludes the proof of Lemma 4.25.

6 Proof of Lemma 4.15

We make the following assumptions in this section.

- To simplify notation, we write $F_i, \mathcal{F}_i, \mathbb{R}_i$ for $\widetilde{F}_i, \widetilde{\mathcal{F}}_i, \widetilde{\mathbb{R}}_i$ this section.
- Assumptions 4.4 and 4.7 run throughout the rest of the paper.
- $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is an L-initial and extremal $(n, k_1, \ldots, k_{s+1} \,||\, k_{s+2}, \ldots, k_t)$-cross intersecting system with IDs $F_1, \ldots, F_t$ under Assumptions 4.4 and 4.7, and for each $i \in [s+1]$, $\{1, n-k_i+2, \ldots, n\} \precneqq F_i \precneqq \{k, n-k_i+2, \ldots, n\}$.

Proposition 6.1. *Let $x \in [r]$, $i \in [\ell_{x-1}+1, \ell_x]$, and let $y \leq x-1$. If for any $j \in [\ell_y+1, s+1]$, we have either $F_j \prec F_i$, or F_j is the k_j-parity of F_i, then*

(i) for any $h \in [\lambda_{\ell_x}+1, \lambda_{\ell_y}]$, F_h is the k_h-partner of F_i;

(ii) for any $h \in [\ell_y+1, \ell_x]$, F_h is the corresponding k_h-set of F_i.

Proof. Let $h \in [\lambda_{\ell_x}+1, \lambda_{\ell_y}]$. By Assumption 4.4 and Remark 4.5, we can see that $\mathcal{F}_i$ and $\mathcal{F}_h$ are non-freely cross intersecting, and for each $j \in [\ell_y]$, $\mathcal{F}_j$ and $\mathcal{F}_h$ are freely cross intersecting. Let $F_{h,i}$ be the k_h-partner of F_i. Note that $i \in [\ell_y+1, \ell_x]$ and for any $j \in [\ell_y+1, s+1]$, we have either $F_j \prec F_i$, or F_j is the k_j-parity of F_i. Since $\mathcal{F}_1, \dots, \mathcal{F}_t$ are cross intersecting, by Facts 2.7 and 2.3, we have $F_h \prec F_{h,i}$. Moreover,

$$(\mathcal{F}_1, \dots, \mathcal{F}_{\lambda_{\ell_x}}, \mathcal{L}(F_{\lambda_{\ell_x}+1}, k_{\lambda_{\ell_x}+1}), \dots, \mathcal{L}(F_{\lambda_{\ell_y}}, k_{\lambda_{\ell_y}}), \mathcal{F}_{\lambda_{\ell_y}+1}, \dots, \mathcal{F}_t)$$

is cross intersecting. Since $(\mathcal{F}_1, \dots, \mathcal{F}_t)$ is extremal, we have $F_h = F_{h,i}$ and get (i). Therefore, for each $h \in [\ell_y+1, \ell_x]$, F_h is lexicographically maximal without exceeding F_i. This implies that for each $h \in [\ell_y+1, \ell_x]$, F_h is the corresponding k_h-set of F_i, we get (ii). □

Proposition 6.2. *For each $i \in [s]$, either $F_{i+1} \prec F_i$, or F_i is the k_i-parity of F_{i+1}.*

Proof. Suppose the assertion of this proposition is false. Then the forthcoming definitions of i, x, j_0 and y are well-defined. Let $i \in [2, s+1]$ with $i \in [\ell_{x-1}+1, \ell_x]$, $j_0 \in [\ell_{x-1}]$ and $j_0 \in [\ell_{y-1}+1, \ell_y]$ be the integers satisfying the following four conditions.

(a) For each $j \in [j_0]$, we have $F_i \supsetneqq F_j$, and F_i is not the k_i-parity of F_j.
(b) For each $j < j_0$, we have either $F_{j+1} \prec F_j$, or F_j is the k_j-parity of F_{j+1}.
(c) For each $j \in [j_0+1, s+1]$, we have either $F_j \prec F_i$, or F_j is the k_j-parity of F_i.
(d) There exists $j \in [j_0+1, i-1]$ such that $F_j \supsetneqq F_i$ and F_j is not the k_j-parity of F_i.

Claim 6.3. $j_0 = \ell_y$.

Proof. If $j_0 = 0$, then $y = 0$, we are done. Suppose that $j_0 > 0$. Suppose on the contrary that $j_0 < \ell_y$. So $\lambda_{j_0} = \lambda_{\ell_y}$. From the choices of j_0 and i, we can see that for any $h \in [\ell_{y-1}+1, s+1]$, $F_h \prec F_{\ell_{y-1}+1}$ and $F_{\ell_y} \supsetneqq F_i$. By Proposition 6.1 (ii), F_{ℓ_y} is the corresponding k_{ℓ_y}-set of $F_{\ell_{y-1}+1}$. Note that $k_i < k_{\ell_{y-1}+1}$ and $F_i \prec F_{\ell_{y-1}+1}$. Thus $F_i \prec F_{\ell_j}$, a contradiction. □

Combining condition (c), Claim 6.3 and Proposition 6.1 (ii), we conclude that

$$F_j \text{ is the corresponding } k_j\text{-set of } F_i \text{ for each } j \in [j_0+1, \ell_x]. \tag{24}$$

Combining Claim 6.3 and Proposition 6.1 (i), we conclude that

$$F_j \text{ is the } k_j\text{-partner of } F_i \text{ for each } j \in [\lambda_i+1, \lambda_{j_0}]. \tag{25}$$

By Fact 2.8, , we conclude that

$$F_j \prec F_{\lambda_i+1} \text{ or } F_{\lambda_i+1} \text{ is the } k_{\lambda_i+1}\text{-parity of } F_j \text{ for all } j \in [\lambda_i+2, \lambda_{j_0}]. \tag{26}$$

Claim 6.4. (F_i, F_{λ_i+1}) *is maximal.*

Proof. Assume on the contrary that (F_i, F_{λ_i+1}) is not maximal. For each $j \in [\lambda_i+1, \lambda_{j_0}]$, let $F_{i,j}$ be the k_i-partner of F_j. Let $h \in [\lambda_i+1, \lambda_{j_0}]$. Then F_h is the k_h-partner of F_i since (25). By Fact 2.6, we have $(F_{i,h}, F_h)$ is maximal, in particular, $(F_{i,\lambda_i+1}, F_{\lambda_i+1})$ is maximal. Since (F_i, F_{λ_i+1}) is not maximal, $F_i \supsetneqq F_{i,\lambda_i+1}$. By (26) and Fact 2.7, we have $F_{i,\lambda_i+1} \prec F_{i,j}$ for all $j \in [\lambda_i+2, \lambda_{j_0}]$. This implies that for each $j \in [\lambda_i+1, \lambda_{j_0}]$, $\mathcal{L}(F_{i,\lambda_i+1}, k_i)$ and $\mathcal{L}(F_j, k_j)$ are cross intersecting. Thus $(\mathcal{F}_1, \dots, \mathcal{F}_{i-1}, \mathcal{L}(F_{i,\lambda_i+1}, k_i), \mathcal{F}_{i+1}, \dots, \mathcal{F}_t)$ is an $(n, k_1, \dots, k_t)$-cross intersecting system. However, $F_i \supsetneqq F_{i,\lambda_i+1}$ implies $\mathcal{F}_i \subsetneqq \mathcal{L}(n, F_{i,\lambda_i+1}, k_i)$, this is a contradiction to the assumption that $(\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_t)$ is extremal. □

Let us continue the proof of Proposition 6.2. Let j be an integer that satisfies condition (d). We claim that $j \notin [\ell_{x-1}+1, \ell_x]$. Indeed, if $j \in [\ell_{x-1}+1, \ell_x]$, then $\lambda_j = \lambda_i$ and $n \geq k_j + k_{\lambda_i+1} \geq k_i + k_{\lambda_i+1}$. So F_i has k_j-parity. (Proposition 2.10 guaranteed the existence of k_j-parity of F_i.) By (24), F_j is the k_j-parity of F_i, a contradiction. Combining this with Proposition 6.1 (ii), we have the following claim.

Claim 6.5. *For each $h \in [\ell_{x-1}+1, i]$, F_h is the k_h-parity of F_i (Proposition 2.10 guarantees the existence of k_h-parity of F_i.) For each $h \in [i, \ell_x]$, F_h is the corresponding k_h-set of F_i (by Claim 6.1 (ii)).*

Since $j < i$ and $j \notin [\ell_{x-1}+1, \ell_x]$, $\lambda_j > \lambda_i$. Combining (25) and Fact 2.6, for each $d \in [\lambda_j+1, \lambda_{j_0}]$, there is a k_i-set $F_{i,d}$ such that $(F_{i,d}, F_d)$ is maximal. In particular, $(F_{i,\lambda_j+1}, F_{\lambda_j+1})$ is maximal. Note that $n \geq k_j + k_{\lambda_j+1} \geq k_i + k_{\lambda_j+1}$. By Proposition 2.10, for each $d \in [\lambda_j+1, \lambda_{j_0}]$, there is a k_j-set $F_{j,d}$ such that $(F_{j,d}, F_d)$ is maximal. In particular, $(F_{j,\lambda_j+1}, F_{\lambda_j+1})$ is maximal. By Fact 2.6, we have $F_j \prec F_{j,\lambda_j+1}$.

Combining (26), Fact 2.8 and Fact 2.7, we get $F_{j,\lambda_j+1} \prec F_{j,\lambda_j+2} \prec \cdots \prec F_{j,\lambda_{j_0}}$. So $(\mathcal{F}_1, \ldots, \mathcal{F}_{j-1}, \mathcal{L}(n, F_{j,\lambda_j+1}, k_1), \mathcal{F}_{j+1}, \ldots, \mathcal{F}_t)$ is an $(n, k_1, \ldots, k_t)$-cross intersecting system. Thus, $F_j = F_{j,\lambda_j+1}$; otherwise, the proper inclusion $\mathcal{F}_j \subsetneqq \mathcal{L}(n, F_{j,\lambda_j+1}, k_j)$ would contradict to the maximality of $(\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_t)$. So F_j is the k_j-parity of F_{i,λ_j+1}. Note that $(F_{i,\lambda_j+1}, F_{\lambda_j+1})$ is maximal and (F_j, F_{λ_j+1}) is maximal. By (24), $F_i \not\succeq F_{i,\lambda_j+1}$. Thus, $F_i \prec F_j$, a contradiction again. This completes the proof of Proposition 6.2. □

Using a similar argument of the proof of Claim 6.4, we obtain the following proposition.

Proposition 6.6. *Suppose that for each $i \in [s]$, either $F_{i+1} \prec F_i$ or F_i is the k_i-parity of F_{i+1}. Then for every $j \in [r]$, $(F_{\ell_{j-1}+1}, F_{\lambda_{\ell_j}+1})$ is maximal.*

Combining Propositions 6.6 and 6.1 (ii), we have the following proposition.

Proposition 6.7. *Suppose that for each $i \in [s]$, either $F_{i+1} \prec F_i$ or F_i is the k_i-parity of F_{i+1}. Then for each $x \in [r]$, and for each $i \in [\ell_{x-1}+1, \ell_x]$, F_i is the corresponding k_i-set of $F_{\ell_{x-1}+1}$.*

We are ready to give the proof of Lemma 4.15.

Proof of Lemma 4.15. We only need to prove that for each $i \in [s+1]$, F_i is the corresponding k_i-set of F_1. Indeed, once this is established, the second part of Lemma 4.15 follows immediately by applying Proposition 6.1 (i).

Suppose on the contrary that $i \in [s+1]$ is the smallest integer such that F_i is not the corresponding k_i-set of F_1. By Propositions 6.2, 6.6 and 6.7, $i = \ell_x + 1$ for some $x \geq 1$, for each $j \in [\ell_x]$, F_j is the corresponding k_j-set of F_1, and for each $y \in [x]$, each $j \in [\lambda_{\ell_y}+1, \lambda_{\ell_{y-1}}]$, F_j is the k_j-partner of $F_{\ell_{y-1}+1}$.

We may assume that F_1 is the first member of $\mathbb{R}_1$ satisfying the conditions of Lemma 4.15.

First, we consider the case: $\ell(F_1) > 0$. In this case, we are going to applying Lemma 4.25 on the tuple $(n, k_1, \ldots, k_{\ell_x}, k_{\lambda_{\ell_x}+1}, \ldots, k_t)$. Since $\ell(F_1) > 0$, we can define F_1' and F_1'' as follows:

$$F_1' = F_1 \setminus \{n - \ell(F_1) + 1\} \cup \{\max(F_1 \setminus F_1^t) + 1\}$$
$$F_1'' = F_1 \setminus \{\max(F_1 \setminus F_1^t)\} \cup \{\max(F_1 \setminus F_1^t) + 1\}.$$

Then $\{1, n-k_1+2, \ldots, n\} \not\succeq F_1' \not\succeq \{k, n-k_1+2, \ldots, n\}$ and $\{1, n-k_1+2, \ldots, n\} \not\succeq F_1'' \prec \{k, n-k_1+2, \ldots, n\}$. For each $j \in [\ell_x]$, let F_j', F_j'' be the corresponding k_j-set of F_1', F_1'', respectively; for each $y \in [x]$, each $j \in [\lambda_{\ell_y}+1, \lambda_{\ell_{y-1}}]$, let F_j', F_j'' be the k_j-partner of $F'_{\ell_{y-1}+1}, F''_{\ell_{y-1}+1}$ respectively. Let $\mathbb{S}', \mathbb{S}''$ be systems obtained by replacing each F_j (where $j \in [\ell_x] \cup [\lambda_{\ell_x}+1, t]$) with F_j', F_j'', respectively. Since $F_{\ell_x+1} \not\succeq F_1$ and F_{ℓ_x+1} is not the corresponding k_{ℓ_x+1}-set of F_1, $\mathbb{S}', \mathbb{S}''$ are $(n, k_1, \ldots, k_t)$ cross intersecting systems. Note that F_1', F_1, F_1'' coincide with A, B, C in Lemma 4.25. We define $\widetilde{f}'$ on the tuple $(n, k_1, \ldots, k_{\ell_x}, k_{\lambda_{\ell_x}+1}, \ldots, k_t)$ to coincide with $\widetilde{f}$ on $(n, k_1, \ldots, k_{s+1}, k_{s+2}, \ldots, k_t)$ of Lemma 4.25. Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, $\widetilde{f}'(F_1') < \widetilde{f}'(F_1)$ (by the choice of F_1) and $\widetilde{f}'(F_1'') \leq \widetilde{f}'(F_1)$. However, Lemma 4.25 gives $\widetilde{f}'(F_1) < \widetilde{f}'(F_1'')$, this is a contradiction, we are done.

Next, we consider the case: $\ell(F_1) = 0$. In this case, let G_1' and G_1'' be two k_1-sets such that $G_1' < F_1 < G_1''$ under lex order. Then $\{1, n-k_1+2, \ldots, n\} \prec G_1' \not\succeq \{k, n-k_1+2, \ldots, n\}$

and $\{1, n-k_1+2, \ldots, n\} \precneqq G_1'' \precneqq \{k, n-k_1+2, \ldots, n\}$. Let d be the integer such that $k_1 = \cdots = k_d > k_{d+1}$. For each $j \in [d]$, let $G_j' = G_1'$ and $G_j'' = G_1''$; for each $j \in [\lambda_1+1, t]$, let G_j', G_j'' be the k_j-partner of G_1', G_1'', respectively. Let $\mathbb{T}', \mathbb{T}''$ be systems obtained by replacing each F_j (here, $j \in [d] \cup [\lambda_1+1, t]$) with G_j', G_j'', respectively. By the choice of ℓ_x+1 and $\ell(F_1) = 0$, we can see that $\mathbb{T}', \mathbb{T}''$ are $(n, k_1, \ldots, k_t)$ cross intersecting systems.

Since $k_1 = \cdots = k_d$ and $n \geq k_1 + k_{\lambda_1+1}$, we may define α', β' and f_1' functions on the tuple $(n, k_1, \ldots, k_d, k_{\lambda_1+1}, \ldots, k_t)$ to coincide with α, β and f_1 functions on $(n, k_1, \ldots, k_t)$ (see definitions (4), (5, (6))). Since $(\mathcal{F}_1, \ldots, \mathcal{F}_t)$ is extremal, $f_1'(G_1') \leq f_1'(F_1)$ and $f_1'(G_1'') \leq f_1'(F_1)$. Since $G_1' < F_1 < G_1''$ and $\max F_1 < n$, $\max F_1 < \max G_1''$. By Proposition 3.3, $\alpha'(G_1', F_1) = \alpha'(F_1, G_1'') = d$, and $\beta'(G_1', F_1) \geq \beta'(F_1, G_1'')$. Thereby, $\beta'(G_1', F_1) = \beta'(F_1, G_1'')$ and $f_1'(G_1') = f_1'(F_1)$. Indeed, if $\beta'(G_1', F_1) > \beta'(F_1, G_1'')$ or $f_1'(G_1') < f_1'(F_1)$, then $f_1'(G_1'') > f_1'(F_1)$, a contradiction. By Proposition 3.3, $f_1'(G_1') = f_1'(F_1)$ implies $d = t - \lambda_1$, and $\beta'(G_1', F_1) = \beta'(F_1, G_1'')$ implies $n = k_1 + k_{\lambda_1+1} = \cdots = k_1 + k_t$. However, by Proposition 4.10[‡], $F_1 = \{k, n-k_1+2, \ldots, n\}$, a contradiction to the choice of F_1. We are done. □

7 Acknowledgements

The authors would like to thank Andrey Kupavskii for valuable comments and suggestions.

References

[1] R. Ahlswede, L.H. Khachatrian, The complete intersection theorem for systems of finite sets, European J. Combin., 18 (2) (1997) 125–136.

[2] P. Borg, C. Feghali, The maximum sum of sizes of cross-intersecting families of subsets of a set, Discrete Math. 345 (2022) 112981.

[3] I. Dinur, S. Safra, On the hardness of approximating minimum vertex cover, Ann. Math. 162 (2005) 439–485.

[4] D.E. Daykin, Erdős–Ko–Rado from Kruskal–Katona, J. Combin. Theory Ser. A 17 (1974) 254–255.

[5] P. Erdős, C. Ko, R. Rado, Intersection theorems for systems of finite sets, Quart. J. Math. Oxf. 2(12) (1961) 313–320.

[6] P. Frankl, A. Kupavskii, Erdős–Ko–Rado theorem for $\{0,1\}$-vectors, J. Combin. Theory Ser. A 155 (2018) 157–179.

[7] P. Frankl, A. Kupavskii, Sharp results concerning disjoint cross intersecting families, European J. Combin. 86 (2020) 103089.

[8] P. Frankl, A. Kupavskii, A size-sensitive inequality for cross intersecting families, European J. Combin. 62 (2017) 263–271.

[9] P. Frankl, N. Tokushige, Some best possible inequalities concerning crossing-intersecting families, J. Combin. Theory Ser. A 61 (1992) 87–97.

[10] P. Frankl, N. Tokushige, Invitation to intersection problems for finite sets, J. Combin. Theory Ser. A 144 (2016) 157–211.

[11] Y. Huang, Y. Peng, J. Wang, The maximum sum of sizes of non-empty pairwise cross intersecting families, J. Combin. Theory Ser. B 178 (2026) 145–163.

[‡]Here, we need to repalce k_t in the proof of Proposition 4.10 by k.

[12] Y. Huang, Y. Peng, Non-empty pairwise cross intersecting families, J. Combin. Theory Ser. A 211 (2025), No. 105981.

[13] Y. Huang, Y. Peng, Complete characterization of maximum pairwise cross intersecting families (I), (2025), arXiv:2601.01929v1

[14] A.J.W. Hilton, E.C. Milner, Some intersection theorems for systems of finite sets, Quart. J. Math. Oxf. 2 (18) (1967) 369–384.

[15] A.J.W. Hilton, The Erdős–Ko–Rado theorem with valency conditions, Unpublished manuscript, 1976.

[16] A.J.W. Hilton, An intersection theorem for a collection of families of subsets of a finite set, J. London Math. Soc. 2 (1977) 369–376.

[17] G.O.H. Katona, A theorem of finite sets, in: Theory of Graphs, Proc. Colloq. Tihany, Akadémai Kiadó, (1968) 187–207.

[18] N. Keller, N. Lifshitz, The Junta method for hypergraphs and the Erdős–Chvátal simplex conjecture, Adv. Math. 392 (2021) 107991.

[19] J.B. Kruskal, The number of simplices in a complex, in: Math. Opt. Techniques, Univ. of Calif. Press, (1963) 251–278.

[20] M. Matsumoto, N. Tokushige, The exact bound in the Erdős–Ko–Rado theorem for cross intersecting families, J. Combin. Theory Ser. A 52 (1989) 90–97.

[21] C. Shi, P. Frankl, J. Qian, On non-empty cross-intersecting families, Combinatorica 42 (2022) 1513–1525.

[22] M. Zhang, T. Feng, A note on non-empty cross intersecting families, European J. Combin. 120 (2024) 103968.